

ORIGINAL ARTICLE

A novel hybrid robust architecture for automatic screening of glaucoma using fundus photos, built on feature selection and machine learning-nature driven computing

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Abstract

Glaucoma is a leading cause of permanent vision loss. Early detection and treatment of this infection is critical for recovery and slowing the progression of vision loss. An efficient novel system focused on customized particle swarm optimization (CPSO) and four state-of-the-art machine-learning classifiers is proposed to boost prediction performance. This interconnected architecture detects glaucoma through five main phases: (1) pre-processing, (2) segmentation, (3) feature extraction, (4) finding the best scored features, and (5) classification using the proposed CPSO-machine learning dependent classifier. The subject images belong to the publically available benchmark Digital Retinal Images for Optic Nerve Segmentation retinal fundus data set. Rather than focusing on the initial 20 extracted features of the retinal fundus, half of the critical features are chosen to form a feature vector based on scores provided by the univariate method and the feature importance method separately. These features are fed into this system for training, testing, and multiple sets of results are created as a result of multiple combinations of CPSO and supervised machine-learning classifiers. These result sets are evaluated using six efficiency metrics. According to the simulation results, the best output is recorded when a univariate selected feature vector is fed into the CPSO-K-nearest neighbour dependent hybrid method. This model outperformed other models with a maximum accuracy of 0.99, a specificity of 0.96, a sensitivity of 0.97, a precision of 0.97, an F1-score of 0.97, and a Kappa of 0.94. A fivefold cross-validation method is used to derive the values. This research would help to achieve good levels of glaucoma care since the proposed system is excellent at distinguishing between stable and glaucomatous eyes. For ophthalmologists, this new technique can be used as second opinion for improving diagnostic accuracy for glaucoma.

KEYWORDS

glaucoma detection, hybrid model, K-nearest neighbour, machine learning, particle swarm optimization

1 | INTRODUCTION

Glaucoma affects millions of people worldwide each year and is a leading cause of blindness. Its development will result in irreversible blindness in people, so it is important that it be detected at an early stage (Roodhooft, 2002). The disorder causes irreversible damage to the optic nerve's central portion, resulting in blindness. Glaucoma is the second leading cause of blindness (13% of all eye-related diseases), and it is characterized

by continuous injury or harm to the optic nerve and a subsequent lack of insight vision (Tuck & Crick, 1992). It is the primary cause of permanent blindness worldwide. In India alone, there were 11.2 million patients with open glaucoma in 2012 (George & Ve Ramesh, 2010). Many published studies estimate that as the population grows, more than 79.6 million people will become sick with this disease (Quigley & Broman, 2006). In the United States, it is estimated that there will be ~7.32 million patients by 2050 (Vajaranant et al., 2012). Some previous published research predicts that this epidemic will infect more than 60 million people sooner or later. As a result, it is important to detect glaucoma early on in order to prevent or at least slow its development and preserve vision (Haleem et al., 2013). Glaucoma-induced changes in the internal structure of the retinal layers must be recognized since early diagnosis and care can avoid vision loss. Identifying these changes from images of the potential patients is currently a hot topic (domain) in the academic community. The overall aim of the research is to build an automatic computer-aided glaucoma identification system that takes retinal fundus images as input and classifies the patient's input images into two classes as output (glaucoma or healthy). This automated system would have many advantages, including the replacement of expensive advanced hardware-based diagnosis machines, the replacement of ubiquitous conventional detection methods, an increase in identification accuracy, a reduction in variability, a reduction in diagnosis time, and, at the same time, it could replace (or at least reduce reliance on) expert ophthalmologists (required for the complex nature of glaucoma disease prediction). The proposed system's efficiency may be improved by embedding it in an IoT computer and/or making it usable in the cloud world. This would benefit people in developing nations or where there is a lack of hardware, medical framework, or highly trained medical practitioners.

Glaucoma develops when the strain in the eye exceeds normal levels and the aqueous fluid does not flow properly. Glaucoma (Figure 1) is a very common condition that involves the optic nerve, the portion of the eye that is described as an electrical impulse. These electrical signals are transmitted to the brains and contribute to vision loss. It is caused by high pressure, that is, intraocular pressure, when the colliery body rate is out of balance with the drainage rate in the drainage pipe. This eye disease may eventually damage the optic nerve. Glaucoma can be discovered only in the interim period to minimize visual damage; afterwards, treatment is difficult. When a patient has had glaucoma for a long period of time, he or she may eventually go blind.

Manual recognition is difficult because it relies heavily on the specialist medical practitioners. Several generous attempts have been made during the last two decades to develop methods for the automatic diagnosis of glaucoma, either with or without the usage of machines. The depth/diameter of the cup (Cup-to-disc ratio—CDR) value is a key measure of glaucoma; if the CDR value is large, it is known as glaucomatous eyes since the CDR rises in the early and even moderate stages. However, CDR is not the only factor that determines whether a person has glaucoma. A range of additional features, such as optic disc diameter, rim depth, retinal nerve fibre layer thickness, and cell count, should be investigated. The optic disc is imaged using two-dimensional stereo microscopy or two-dimensional colour fundus imaging. Glaucoma can only be used in images by using CDR processing to distinguish between an infected eye and a stable eye (as shown in Figure 1).

For glaucoma screening, funduscopy, and optical coherence tomography (OCT) are used as two traditional imaging modalities. When compared with OCT, funduscopy is easier to employ and provides an exaggerated view. However, it has the potential to imprison the structural alterations in the outermost retina layer that result in the manifestation of the untreatable and final stage (de Sousa et al., 2017). A retinal camera is used for fundus photography, which grasps the optical nerve, choroid, and the retina. This camera, also known as a fundus camera, captures a 3D retina image as a 2D image. Colour fundus imaging uses natural light to capture the image of the retina. These captured fundus images consist of vessel segment and optic disk, which are segmented and continuously monitored to help the experts recognize disorders present in the patient's eye (Gajbhiye & Kamthane, 2015). To capture the fundus image of the patient's eyes, the eyes are first enlarged to increase the observation angle so that the image can capture a larger portion of the eye. Fundus images of the public benchmark data set are used in the proposed work. In the case of the presence of glaucoma, the vessel segment is to be keenly observed because it shows an incessant variation. An example of a fundus image of the left and right eye is shown in Figure 2.

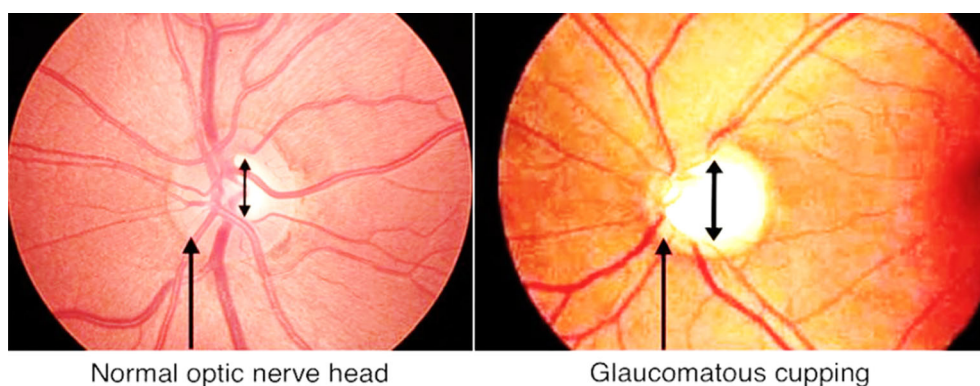


FIGURE 1 Normal eye (left) and glaucoma infected eye (right)

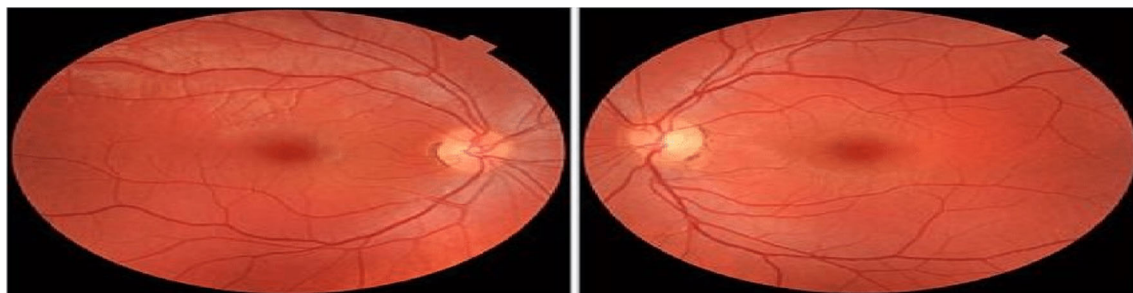


FIGURE 2 Left eye image and right eye image

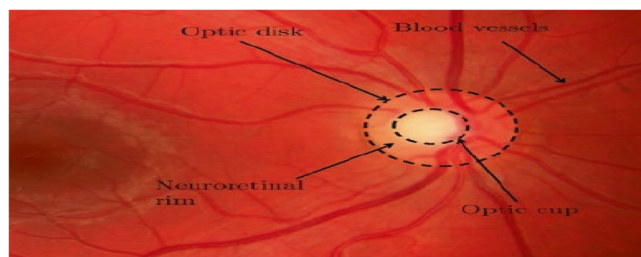


FIGURE 3 Parts of eye visible in fundus image



FIGURE 4 Left image is of the glaucomatous eye, and the right image is of the normal eye

In Figure 3, one can observe the different portions of the fundus image required to diagnose glaucoma with their description as follows:

1. Optic disk: It is an area that is elliptical in shape that is brighter than its neighbouring region at the centre of fundus image.
2. Neuroretinal Rim: It is the region between the optic disk and the optic cup.
3. Optic Cup: The optic cup is the region in the middle of the optic disc that is white, cup like.
4. Cup disk ratio: It is the ratio of the size of the optic cup to the optic disk.

It is clear from Figure 4 that in the glaucomatous eye, the size of the optic cup increases, which leads to an increase in cup to disk ratio.

The purpose and objective of the following paper are mentioned below.

1. The objective is to investigate the global trend in new directions like the inferior region, superior region, nasal region, and temporal region of the human retina. This entails the practice of interpreting the eye's basic structure.
2. The aim of the study is to include a comparative analysis of various machine learning algorithms (support vector machine—SVM, K-nearest neighbour, random forest, and decision tree) dependent self-adapting models for sensitivity, specificity, and the other five metrics.

3. To evaluate the performance of the feature vector, which comprises the features returned by the 'univariate' method and the 'feature importance' method. We are also determined to identify the number of features required for high classification accuracy. To evaluate the performance of the customized particle swarm optimization (CPSO) and to analyse the efficiency of the features returned by it in terms of accuracy (and other parameters) achieved.
4. To analyse and observe the performance of the set of results generated by the proposed three-layered architecture (statistical-nature inspired-machine learning) for glaucoma identification and to propose the best approach for the research fraternity. To justify the superiority of the presented work by comparing with other state-of-the-art published work.
5. Thus, this work offers a fast, efficient, and automated glaucoma identification system based on a combination of bio-inspired computing approaches and machine learning approaches. A more accurate approach is recommended for glaucoma diagnosis. The proposed computerized automated system for early and timely glaucoma identification requires the least amount of human intervention with high proficiency. The study fills up the research gap by proposing a system based on artificial intelligence approaches and outputs with high accuracy standards that are on par with various ophthalmologist societies.

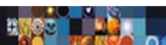
The rest of the paper is organized as follows. In the introduction section, the subject of glaucoma and its patients are introduced, the statistics concerning this disease are shared, the need for further research is explained, the requirements of the researcher's community are mentioned, and the usability of the researcher and the public's benefits from this research are described. Later in this section of the paper, there is a discussion on fundus images and a contrast between the normal eye and the glaucoma-infected eye. In Section 2, a detailed discussion of previous studies published on glaucoma identification from retinal fundus images has been presented. In Section 3, the following outline is given for the proposed data gathering methodology: (first part) features to be extracted for diagnosis: How to calculate feature values (second part). At the same time, a detailed note on the proposed system is also conveyed in the body of the document. Section 4 details the results of studies and the performance measurements of different machine learning algorithms for the mentioned purposes. A comprehensive comparative analysis with previously published prominent studies is also shared. Section 5 is dedicated to discussion of the whole work. Section 6 depicts the real application of the presented work. Conclusion, Section 7, is located at the end.

2 | PRIOR PUBLISHED STUDIES

In this section, review and examination of several notable published papers have been done, including the most recent reputable articles, in which researchers focused on glaucoma detection using retinal imaging.

The paper suggested by Al-Bander et al. (2017) explores the possibility of using deep learning techniques [convolutionary neural networks (CNNs) architecture] to achieve an integrated system for learning features for glaucoma detection. CNN automatically removes attributes from raw images and transfers them to the SVM. Raghavendra, Bhandary, et al. (2018) proposed a glaucoma screening protocol in which they initially performed the segmentation of the optic disc and, after that, they performed the glaucoma detection radon transformation technique. A collection of non-parametric (dynamic) and glaucoma-specific (i.e., non-global image descriptor) properties are extracted from fundus images to scan for glaucoma. Phan et al. (2019) used CNNs to use colour fundus images to distinguish two glaucoma locations and the variables based on their discriminative ability. The author's project is of high quality and a benchmark image improves this infection diagnosis. Although it was not clearly specified, the suggested solution was to be paired with an artificial neural network (ANN) to generate patterns. Khalil et al. (2017) proposed a computer-based glaucoma recognition method consisting of two hybrid textures feature-set (HTF) and hybrid structural feature-set (HSF) modules, which distinguish glaucoma from the grouping of fundus and OCT images with the help of a hybrid structural and textual feature. The HSF module helped to define an image using SVM, and for decision-making, the HTF module evaluated the images using various strength and texture-based features along with SVM. Salam et al. (2016) used mixed structural features (such as CDR) and non-structural features (such as texture and strength) in the next segment of the analysis to improve the diagnostic process of glaucoma. The machine-learning module used the results from the study to make its decisions. With machine learning hybridization and structural changes, the efficiency of the model is more noticeable and accurate. Discriminatory criteria and adaptive thresholding strategies were implemented in the analysis for glaucoma diagnosis, which was suggested by Issac et al. (2015). The applied methods extract features like the CDR, the neuro retinal rim (NRR) region and blood vessels for better segmentation. They proposed many ways of obtaining structural traits. The authors recommended using SVM and ANN classifiers to diagnose retinal images. The NRR, CDR, and inferior superior nasal temporal (ISNT) vasculature were extracted for categorization. Acharya et al. (2015) offered the glaucoma identification method utilizing Gabor transformation. The high-level classification method removes the top few components of the features and then constructs a data classifier utilizing the well-performing features.

Vijapur and Kunte (2017) suggested that inside glaucoma one needs to look at the thickness of the neuroretinal rim. The rim region and the rim region, when taken together, exhibited a tertiary biomarker collection. The authors noticed a substantial increase in the estimates of sensitivity and precision in the grouping. This approach suggests extracting high-dimensional function vectors followed by the compression of the resultant vectors via *principal component analysis* (PCA) and finally the classification of the resultant vectors via SVM. Juneja et al. (2019) suggested



using guided learning to detect glaucoma from non-glaucoma by using a lattice classification method. A technique of implementing a system of pre-processing of photographs to reduce outliers was explored in the report. A moldmaker will need to continue the method of moulding one of the eyes, through which one will be able to extract an offending section of the mould later on, or a portion of the mould that will be changed. García et al. (2020) suggested an approach for the identification of glaucoma from raw spectral domain OCT (SD-OCT) quantities. Long short-term memory networks were employed to aggregate the latent space's recurrent dependencies, resulting in a holistic function vector formed by the suggested sequential weighting module. The article proposes a more accurate method of glaucoma diagnosis utilizing image channels and wavelet processing from fundus pictures. Kirar et al. (2020) proposed an improved approach to diagnose glaucoma; the images were resized in order to guarantee that they were of the same dimension. Then, the segments were removed from the red, green, blue, and grey scale (Gs). In second-stage discrete wavelet transform (DWT), four forms of images are enhanced and decomposed separately into sub-band images (SBIs). Thirdly, only the most useful functionality was omitted from each DWT-SBI. Fourth, attributes originating from red channel (RC), green channel (GR), and blue channel (BC) were concatenated and normalized from the photos of RC, GC, and BC. Finally, these robust features were entered into a classifier and the performance of this classification evaluation (LS-SVM) was finally evaluated. The C-RGBGs-IF approach outperformed the others.

Saxena et al. (2020) introduce the framework for the precise identification of glaucoma based on deep learning using CNN. CNN offers a formal framework for the separation of images. The proposed work for a total of six layers was tested. The dropout mechanism was often used to achieve appropriate efficiency in the diagnosis of glaucoma. Martins et al. (2020) proposed a computer-aided diagnostic (CAD) pipeline capable of using fundus images to diagnose glaucoma and operate offline on mobile devices. Several public fundus image data sets were combined and used to create a system that performs tasks of segmentation and classification. These networks were then used to create a glaucoma evaluation pipeline, which was implemented into a smart phone application in order to test the efficiency of this process in a restricted environment, and time and space complexities were measured. (Abdel-Hamid, 2020) suggested a wavelet-based glaucoma recognition approach for real-time screening systems. A combination of wavelet-based mathematical and textural characteristics was used to decide whether the retinal picture was intact or glaucoma-prone. They found various methods that each gained importance for the diagnosis of glaucoma. The discovered optic disc area's wavelet statistical and textural properties were integrated. The high resolution data set had an accuracy of 96.70% and an area under curve (AUC) of 94.70%. Three wavelet-based statistical and textural analysis strategies were beneficial in glaucoma diagnosis. The proposed study aims to automatically diagnose glaucoma based on fundus pictures (Gour & Khanna, 2020). It detects the global image descriptor (GIST) and pyramid histogram of the pyramid histogram of oriented gradients (PHOG) features in pre-processed fundus photos. The extracted features are ranked and selected using PCA. Images were grouped into Drishti-GS1 and HRF using the SVM classifier. The picture acquisition process includes the transformation of the RGB fundus image into grey and the increase in the contrast of the fundus characteristics (Qureshi et al., 2020). During the extraction process, the boundary between the optical disc and cup was segmented. A CDR was determined to detect glaucoma in the image.

Kanse and Yadav (2020) found a way to integrate both the harmonic operator and the genetic algorithm (GA) to construct their glaucoma classifier, which allows use of both the harmonic operator and the GA as training stimuli. In a recent report, researchers (Shehryar et al., 2020) used both a fundus examination (binocular ophthalmoscopy) and an extremely sensitive OCT scanning system for the eye that explored the back of the eye to determine the hard and soft tissues of the eye. The researchers (Ajesh & Ravi, 2020) used artificial intelligence and machine learning to create a successful approach for glaucoma research. The proposed Jaya-chicken swarm optimization (CSO) was designed to incorporate into the recurrent neural network (RNN) classifier two techniques: the Jaya algorithm, in which the topology of the RNN was created, and the CSO, in which a group of virtual CogNets were trained together to boost their weights in the RNN. In another recent paper on the diagnosis of glaucoma (Ajesh et al., 2020, practitioners wanted to use several features to identify microscopy images. Therefore, they used a machine-learning method and the Daubechies wavelet. The Deep belief neural network (DBN) and DWT classification approaches were introduced and refined to categorize diseases according to treatment options. For the intention of fast tracking of glaucoma, researchers suggested a method focused on a visual examination of fundus images by the use of colour processing and certain other computer vision techniques (Karmawat et al., 2019). The images were collected through PCA for clustering into two clusters. The approach is based on the fuzzy c-means (FCM) algorithm and is used to segment the optic cup. Fundus pictures were classified into normal and glaucoma classes using SVM and ensemble classifiers based on the fusion of segmentation-based and global image-based characteristics. This study used quasi-bivariate variational mode decomposition (QB-VMD) for glaucoma prediction (Kirar et al., 2021). Using QB-VMD, the pre-processed fundus pictures were first decomposed into five SBIs. SSQB-VMD extracted and decomposed just the green channel. The first stage's high-frequency fifth SBI was decomposed into five SBIs using QB-VMD. For glaucoma prediction, authors use a hybrid technique (using DWTs and EWTs) to examine fundus image properties (Kirar & Agrawal, 2018). DWT separates pictures into approximation and detail coefficients, whereas empirical wavelet transforms (EWT) divides them into sub band images. This approach included all DWT and EWT properties. The green channel picture from fundus images was mixed with characteristics derived from each of the four methods: DWT, EWT, DWTEWT, and EWT DWT. Seventy-one characteristics were standardized, sorted, and decomposed and they finally used 14 effective characteristics.

In this study, QB-VMD using digital fundus images was used to automate glaucoma diagnosis (Agrawal et al., 2019). The QB-VMD approach provided band-restricted SBIs concentrated around a certain frequency. A QB-VMD SBI yielded 70 features. The ReliefF approach was used to choose 45 recovered features. Singular value decomposition was used to reduce the dimensionality of certain attributes. The reduced features were SVM classified using least squares. An ECNet-based feature extraction method for identifying glaucoma in fundus images (Nayak

et al., 2021). There are many layers that may assist in extracting discriminating characteristics. They employed an RCGA to improve layer weights. The best performing model was determined from the final feature vectors using classifiers like KNN, and many more. The most accurate SVM model was created by ECNet (97.20%). This research used a machine-learning algorithm and a DWT to classify diseases based on their treatment options. To construct an automated glaucoma detection system, higher order spectra (HOS) cumulants produced by the Radon transform (RT) were used (Noronha et al., 2014). For automated diagnosis, SVM and Naive Bayes (NB) classifiers were given clinically significant linear discriminant (LD) characteristics, and then LDA was used to minimize the number of features.

A blood vessel detection algorithm was used to accurately segment the optic disc (OD) and optic cup (OC) (Pathan et al., 2021). The proposed solution uses a decision tree classifier and the circle finding approach. The suggested OC segmentation approach creates a new channel owing to lower OD/OC pixel variability. They had clinical and textural properties. They employed glaucoma domain knowledge and colour and texture statistics to extract features. An AdaBoost ensemble classifier with dynamic selection was constructed to categorize glaucoma fundus images. After a glaucoma-specific pre-processing step, researchers (Bock et al., 2010) indicated that an appearance-based dimension reduction approach compresses numerous general feature categories. A probabilistic two-stage classification technique was used to derive the Glaucoma Risk Index (GRI). The authors created an automated glaucoma classifier from the retinal fundus images (Abad et al., 2021). The new classifier was tested on a massive collection of retinal fundus images. To test the proposed classifier in a clinically relevant situation, the authors merged all available retinal fundus imaging data sets. The authors present an offline interpretable computer aided diagnosis (CAD) pathway for diagnosing glaucoma using fundus images. This work presented a non-parametric global image descriptor (GIST descriptor) and optic disc localization for glaucoma diagnosis (Raghavendra, Fujita, et al., 2018). After the Radon transformation, unique area-based optic disc segmentation occurs. The census transformation adjusts for the Radon image's light level fluctuation. The spatial envelope energy spectrum was derived from pictures using the GIST descriptor. The rating characteristics were utilized to build a glaucoma classifier. A machine learning-based glaucoma decision-making system was presented (Acharya et al., 2015). They got variance, the mean, kurtosis, and skewness. Finally, SVM and NB were used to classify the data. The segmented optic disc vessels were replaced with medians. Wavelet features for glaucoma detection, PCA, and evolutionary attribute selection were used to identify key characteristics. Classification was done using random forest, naïve bayes, ANN, and SVM. PCA with SVM exceeds others with 94.7% accuracy. The paper (Orlando et al., 2020) explored glaucoma classification and segmentation difficulties. As part of REFUGE, they made accessible 1200 fundus images with ground truth segmentations and clinical glaucoma labelling. To encourage the development of new approaches, they devised an evaluation framework to compare models. The best teams presented discoveries that complemented the ground truth annotations and could be used to assemble the data. Researchers integrated structural and non-structural data to generate a hybrid glaucoma diagnostic feature set (Salam et al., 2016). The CDR, NRR area, and ISNT were obtained. Texture, wavelets, and local binary patterns were also retrieved. The SVM correctly classified normal and abnormal cases.

From the review of the literature on disorders associated with glaucoma, it can be concluded that diagnosing glaucoma is critical in determining the source of the retinal condition that may contribute to optic nerve damage and result in slow and irreversible vision loss. Behind the issue of glaucoma is a critical need for a global dynamics features-dependent model that can be used to evaluate retinal images and aid the medical practitioner in determining glaucoma with the greatest accuracy possible. This study is planned and delivered in the same manner.

3 | EXPERIMENTAL SETUP

This section focus wills on features extracted from retinal fundus images, standard data set used, system (framework) designed for this study and the Customized PSO designed and its implementation in this empirical study.

3.1 | Pre-processing

In the proposed technique, it has been established and determined that the layout of the data (e.g., features) would have been pre-processed before the data extraction stage. For the diagnosis of glaucoma, optic disc, and optic cup area in retinal fundus images is very significant region. The optic disc is the major region of concern in retinal fundus picture where the position of optic cup is at the centre of optic disc. For more processing originally, the image of the retinal fundus has been cropped by $230 \times 230 \times 3$. The optic disc is usually at the middle of the optic disc in the image of the portion of the retina. Figure 5a displays the colours of Red, Green, Blue (RGB) in fundus pictures, which are commonly recognized as Red, Blue and Green display formats for three networks. The blue pattern might be difficult to conduct a number of operations on it. In Figure 5c, the grey picture is seen and it is being used to represent the crucial component of human's eyes and it carries a huge amount of knowledge. First, fundus imaging takes a person's eye scan and transforms into a person's 'grey-green' colour-box. The equalization represents the feature of transformation or conversion that is entirely used to create an output image with a standardized histogram. It is very efficient to change the brightness of the process of equalization of the input picture histogram. The amplitude of the input picture is uniformly distributed around the image as updated. The key emphasis of this technique is the application of strength similarly to the total picture. The linear trend to the cdf (cumulative probability function) is given by this.

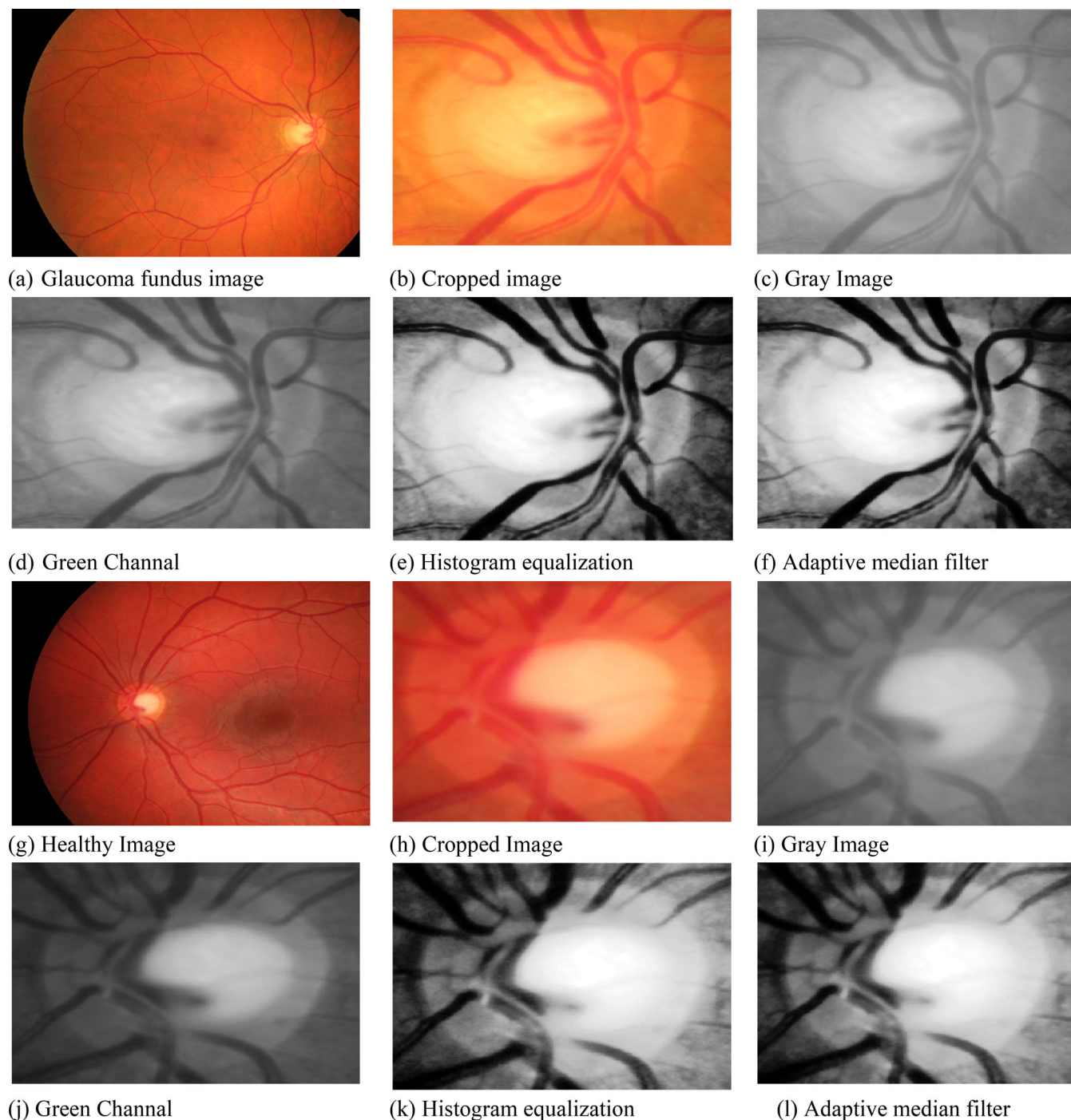


FIGURE 5 (a)–(f) Pre-processing of glaucoma image and (g)–(l) pre-processing of healthy image

In addition, an adaptive median filter has been used to equalize the signal. The picture of fundus may be distorted by pulse noise. Due to electromagnetic interference, impulsive noise is made. In the spatial domain, the adaptive median filter is also an effective filtering strategy to reduce the noise of the impulse. To distinguish the noisy pixels, it contrasts each pixel with its adjacent neighbouring pixels. The Adaptive Median Filter conducts spatial analysis to assess which pixels have been influenced by impulse noise in an image. Through contrasting each pixel in the picture to its adjacent neighbouring pixels, the Adaptive Median Filter classifies pixels as noise. The scale of the neighbourhood, as well as the reference threshold, is customizable. In addition to being not structurally consistent with the pixels, a pixel is known as impulse noise, which varies from any of its neighbours. These noise pixels will then be substituted by the median pixel value of the surrounding pixels that passed the noise marking examination. The pre-processed glaucoma image is shown using Figure 5a–f meanwhile pre-processed healthy image is shown in Figure 5g–l.

3.2 | Optic disc and optic cup segmentation

The field of interest from the input fundus colour images is first clipped to detect the optic cup and optic disk. In this work, the colour image essentially has three wide red, green, and blue channels, except the green channel for further processing. An adaptive median filter has been used in order to eliminate the unnecessary signal from the fundus image as the next step. Histogram equalization is used to include the power distribution in the fundus image in a related manner. It also offers the highest picture quality. Moreover, morphological multistep closing operations have been used to remove unnecessary blood vessels. A method for blob detection in images or thresholding is Maximally Protected External Regions (MSER). The complete pixel below is known as black, and in the fundus image, all those indicated above or equivalent is white. The MSER principle is related to the image's single component tree. During the suggested procedure, MSER is used twice. First, it is used to find the disc in the fundus image, and then it is used to separate the cup zone from the disc area.

3.3 | Feature extraction

The field of interest from the input fundus colour images is first clipped to detect the optic cup and optic disk. Basically, the colour images are used to fetch features from a large image database containing critical information (Singh & Garg, 2020). In this empirical study, 20 characteristics were obtained from the retinal fundus image array: similarity, homogeneity, *SD* disk, comparison, energy disc, mean disc, entropy disc, entropy cup, mean cup, area disc, energy disc, radius disc, cup-to-disc ratio, energy cup, area disc, area cup, temporal region area, inferior region area, nasal region area, and superior region area. A function extraction index is created by this set of features.

3.3.1 | Homogeneity

It is possible to determine the distribution of grey or colour values within an image by using a homogeneity factor. Homogeneity is defined as the state or quality of having an even structure or a similar kind or composition throughout a whole system (Singh et al., 2021). It can be calculated as following formula

$$\sum_{i,j=0}^{N-1} \frac{p_{ij}}{1 + (i-j)^2}, \quad (1)$$

where N is representing the number of grey level in the image. Here p_{ij} is representing the element i, j of the normalized symmetrical grey level co-occurrence matrix.

3.3.2 | Contrast

It is the difference between brightness and colour of the object and other objects within the same region (Singh et al., 2021). When it comes to images, contrast can be simply defined as the difference between the maximum and minimum pixel intensities. It can be calculated as:

$$\sum_{i,j=0}^{N-1} p_{ij}(i-j)^2. \quad (2)$$

Here i and j represent the difference of maximum and minimum intensity in the image. Here p_{ij} is representing the element i, j of the normalized symmetrical grey level.

3.3.3 | Correlation

The presence of a causal relationship can be indicated by the presence of a correlation (on occasion). This function tests the association of a pixel over the entire image with its neighbour.

$$\sum_{i,j} \frac{(i-\mu_i)(j-\mu_j)p(i,j)}{\sigma_i\sigma_j}, \quad (3)$$

Where μ is representing the mean of the all pixel in image. $\sigma_i\sigma_j$ is representing the variance of the intensities of all reference pixels in the relationships.



3.3.4 | Optic cup-to-disc ratio

It is one of the significant features used for detecting glaucoma. One or many cup-to-disc ratio's (CDRs) can be noticeable like horizontally, vertically in the same fundus image. CDR is 0.5 for regular healthy eye and over this ratio, the eye is expected to be glaucoma affected (Qureshi et al., 2020).

$$CDR = \left(\frac{\text{cup area}}{\text{disc area}} \right). \quad (4)$$

Here in Equation (4) cup area is representing the area of the cup region and disc area is representing the area of disc region in retinal fundus image.

3.3.5 | Optic disc mean

It is the mean of segmented optical disc image $F^0_i(l_1, l_2)$ with ROI images

$$M_{d0} = \sum_{l_1=1}^a \sum_{l_2=1}^b \frac{F^0_i(l_1, l_2)}{S^0}, \quad (5)$$

where (l_1, l_2) represents the pixel location in image. Here S^0 shows the total number of pixels in a segmented optical disc.

3.3.6 | Optic cup mean

The result provided by the mean is average of the pixel of segmented optic cup area $F_i(l_1, l_2)$. Here M_0 is the mean of optic cup

$$M_0 = \sum_{l_1=1}^M \sum_{l_2=1}^N \frac{F_i(l_1, l_2)}{T^0}, \quad (6)$$

where (l_1, l_2) represent the pixel location of image. Here T^0 represent the total number of pixel in the image.

3.3.7 | Optic disc entropy

The input image's texture can be described statistically using entropy, a measure of randomness. Entropy features have been extracted for optic disc segmented image $F^0_i(l_1, l_2)$ with the help of following constitute equation:

$$E_0 = - \sum_{a_1=0}^{l^0-1} q_h(a_1) \log_2[q_h(a_1)] \quad (7)$$

Histogram probability is represented by $q_h(a_1)$ for the segmented optic disc image and it is represented by

$$q_h(a_1) = \frac{T^1}{T^0}, \quad (8)$$

where T^1 is total number of pixel at the grey level a_1 .

3.3.8 | Optic cup entropy

A measure of image information content, known as entropy, which is interpreted as the average uncertainty of the information source, can be found here. Entropy is defined in Image as the ability of individual pixels to change their intensity level. Using the entropy value provides a better

comparison for quantitative analysis and evaluation of image details. Entropy features have been extracted for optic disc segmented image $F_i(l_1, l_2)$ with the help of following constitute equation:

$$E_0 = - \sum_{b_1=0}^{l_1-1} q_h^0(b_1) \log_2 [q_h^0(b_1)]. \quad (9)$$

Histogram probability is represented by $q_h^0(b_1)$ for optic cup region at the grey level b_1 .

$$q_h^0(b_1) = \frac{T^{11}}{T^{00}}, \quad (10)$$

where T^{11} is total number of pixel at grey level b_1 . T^{00} is total number of pixel in segmented optic cup region.

3.3.9 | Optic disc energy

The energy of the optic disc area is a measure of how much it has changed in a specific area. Energy features are calculated for the segmented optic disc $F_i(l_1, l_2)$ with the help of following equation:

$$Energy = \sum_{l_1, l_2} F_i^0(l_1, l_2)^2. \quad (11)$$

3.3.10 | Optic cup energy

The energy of the optic cup area is a measure of how much it has changed in a specific area. Energy features are calculated for the segmented optic disc $F_i(l_1, l_2)$ with the help of following equation:

$$Energy = \sum_{l_1, l_2} F_i(l_1, l_2)^2. \quad (12)$$

3.3.11 | Optic disc SD

There are two ways to measure SD: During a time when SD is near zero, the measured values are all close to the mean and all coming together. SD is also high when the values or data are spread out and also far from the mean, which makes it hard to get a good picture of what is going on. Therefore, when the SD is 0 and there are no differences between two numbers, they are the same as each other. The following Equation (13) is used to calculate the SD for the optic disc region.

$$SDD = \sqrt{\frac{\sum_{i=1}^{T^0} (a_i - \bar{a})^2}{T^0 - 1}}. \quad (13)$$

Here a_i represent the pixels in the optic disc image and T^0 is the total number of pixels.

3.3.12 | Optic cup SD

Using the SD of an image means that the image is different. The SD is a way to show how different this is. The SD for optic cup region is computed as with following equation.

$$SDC = \sqrt{\frac{\sum_{i=1}^{T^1} (b_i - \bar{b})^2}{T^1 - 1}}. \quad (14)$$

Here, b_i represent the pixel in optic cup region and T^1 represent the total number of pixels in optic cup.

3.3.13 | ISNT regions

As shown in Figure 6, ISNT stands for different regions, especially I (inferior region), S (superior region), N (nasal region) and T (temporal region), respectively. The ISNT rule is commonly used to distinguish between eyes effected by glaucoma and healthy eyes (Issac et al., 2015). The value of these regions is already established for the normal eye as: inferior region area > superior region area > nasal region area > temporal region area (Singh et al., 2021).

The ISNT regions are crucial in the proposed methodology, since there is a requirement to improve the current model(s) based on image analysis. In order to diagnose glaucoma, the extracted features directly dependent on the area can be considered: inferior, superior, nasal, and temporal in retinal fundus images. The output measurement for each region is shown on a black and white scale in Figure 7 after extracting the ISNT area. This action is carried out on both forms of data sets, that is, pictures with/without glaucoma disease. Usage of the smaller training data at the training level limits the precision of the classification. Therefore, it is important to use more features (Zilly et al., 2017), which is why an adequate number of 20 critical features have been gathered.

3.4 | Features short listing

The data characteristics that have been used to train suggested models of machine learning have a significant effect on performance. Model efficiency may be adversely impacted by unrelated or partly related characteristics. With irrelevant features present in the data, model accuracy can be decreased and suggested models can be learned on the basis of irrelevant features. Thus, two feature selection strategies have been used to boost the efficiency of the presented models. Feature selection is the mechanism in which one picks certain features that add most to a forecast variable or output in which someone is involved, either automatically or manually. In this analysis, the algorithms for *univariate* feature selection and *feature importance* are used. Univariate selection is a test based on statistics that can be used to pick certain characteristics that are most closely related to the output variable. The Scikit learning library includes the Select K Best class that can be used for choosing a certain number of features with a suite of different statistical tests. A feature of importance in machine learning is a feature discovery technique. The technique used for practical relevance is the scoring process. Each input function provides a score. It is also possible to assess which function is more important for predicting the goal based on the ranking. For implementing feature selection, feature importance scores are used.

3.5 | Proposed framework in detail

Retinal fundus photographs belonging to a public database are pre-processed using various mentioned processes for consequent processing. The proposed framework has been investigated and evaluated on the publicly available data set of colour retinal fundus images. The data set used is DRIONS-DB, which provides the ground truth of optic-disc. The database of digital retinal images for optic nerve segmentation (DRIONS-DB) comprises 110 digital retinal images with OD markers created by two doctors. The photos were taken of individuals between the ages of 53 and 54, with 46.2% of male and 53.8% of female patients. This data set contains photos of the retina with two different forms of pathology: 23.1% of images are glaucomatous, and 76.9% are of ocular hypertension. The photographs were captured with an analogue colour backdrop camera,



FIGURE 6 Different area in inferior superior nasal temporal regions

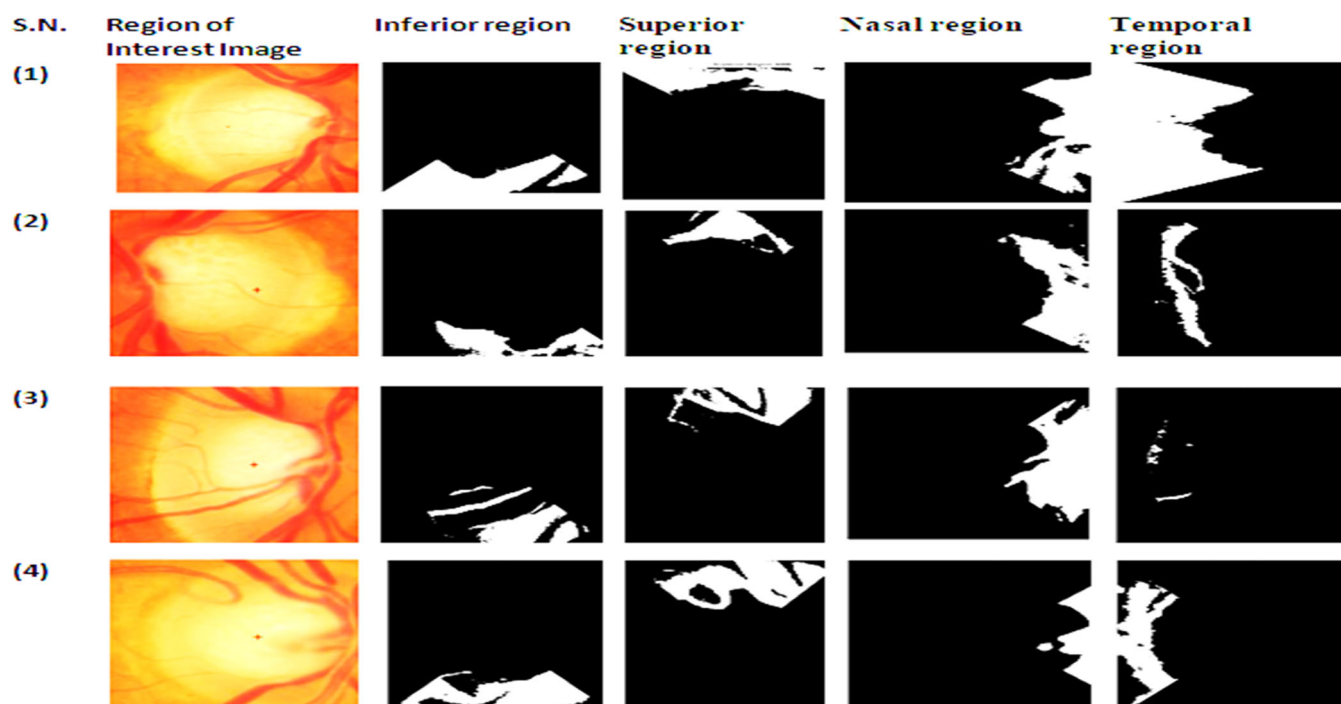


FIGURE 7 Image of the inferior superior nasal temporal (ISNT) regions (extracted from the original image)

recorded on slides, and then scanned using an HP-PhotoSmart-S20 high quality, RGB format, 600×400 resolution scanner. The experimentation is performed with a fivefold cross-validation approach. The code is implemented in the Python language.

Twenty vital features are extracted from these ones. Rather than working on all the extracted features, a shortlist of $\sim 50\%$ of the best-scored features takes place. In order to find the best-scored nine features, two methods have been preferred; 'univariate' and 'feature importance'. These methods return the nine best-scored (significant) features according to their procedure. When the univariate method generates a feature vector with nine features, it sends it to four machine learning-based classifiers. The same thing happens when the feature importance method generates an important-features vector with nine features.

Next, these features are passed on to CPSO. CPSO, which is described in more detail in the next subsection, does this. The ultimate objective of the CPSO is to find the optimized subset of features that can generate the maximum accuracy metric with the support of these four classifiers. Two verticals are propagated from the eight hybrid combinations: four through univariate features (CPSO+KNN, CPSO+SVM, CPSO+Random Forest, and CPSO+Decision Tree) and four through the feature importance method (CPSO+KNN, CPSO+SVM, CPSO+Random Forest, and CPSO+Decision Tree). In total, 16 sets of generated results from the proposed framework are compared on six efficiency metrics to find out the best option for the efficient diagnosis of glaucoma.

3.5.1 | Detailed discussion on CPSO

It is evident and proved from the previous published prominent studies that particle swarm optimization (PSO) is capable of effectively solving diverse real-life single-objective or multi-objective optimization problems (Sakri et al., 2018). In prior published reputed studies (like Sharif et al., 2020), practitioners have proved the applicability of PSO in solving medical (or medical imaging) domain related problems (Zomorodi-moghadam et al., 2021). In this case, the problem at hand belongs to a category of problems where the objective is to find the nominal subset of features from an initial vector of nine features, which maximizes the accuracy metric. In other words, the ultimate objective is the maximization of the accuracy parameter returned by machine learning algorithms when a nominal subset of features created by the proposed CPSO (Figure 8) is directed to four selected machine learning classifiers turn-by-turn. As discussed in the previous section, eight combinations are possible, so eight sets of results will be generated, which will be evaluated on six efficiency evaluation metrics. The pseudocode of the proposed hybrid CPSO-Machine Learning classifier-based algorithm is presented below.

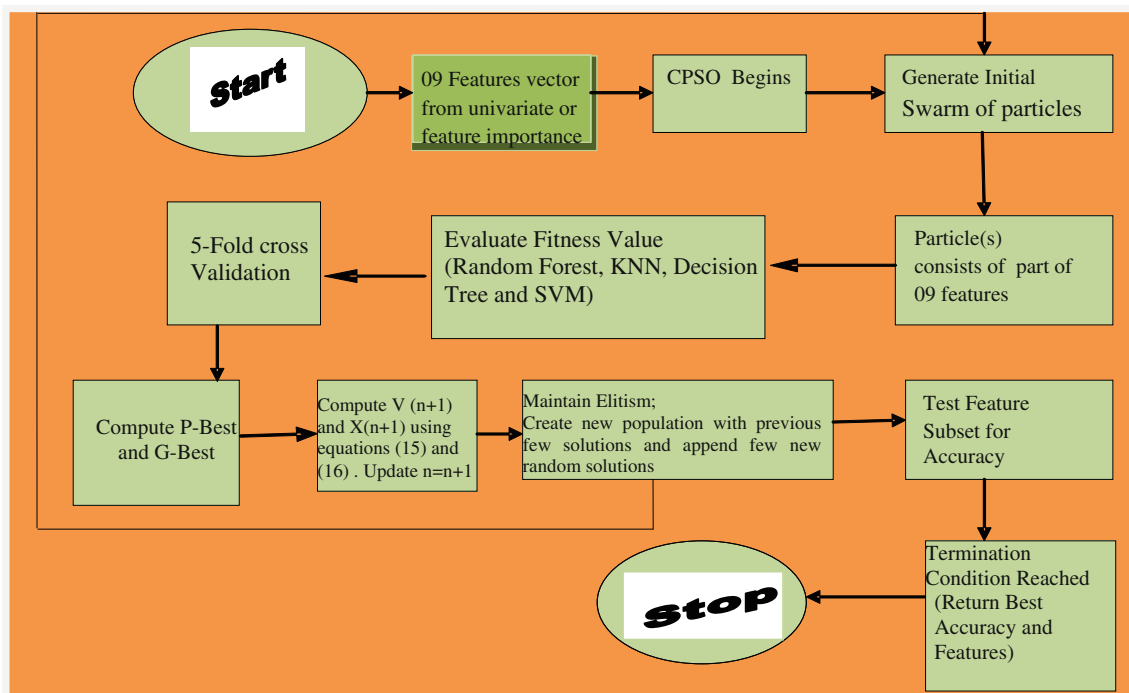


FIGURE 8 Block diagram of implemented hybrid CPSO-machine learning model for glaucoma diagnosis

The basic steps of PSO algorithm are as follows:

Begin (Initialization)

Generate initial swarm $X(0)$
Generate initial velocities $V(0)$

End

Set $n = 0$ (n is the initial number)

Repeat

Begin

Compare fitness value for each individual of swarm

End

Compute $PBest(n)$ and $GBest$

Begin(Perform PSO operations)

Compute $V(n+1)$ from Equation (15)

Compute $X(n+1)$ from Equation (16)

End

Set $n = n + 1$

Until termination criteria satisfied.

New velocity is computed using Equation (15).

$$v_i(k+1) = w * v_i(k) + c_1(p_i - x_i(k)) + c_2(G - x_i(k)) \quad (15)$$

New position is calculated using Equation (16).

$$x_i(k+1) = x_i(k) + v_i(k+1) \quad (16)$$

where, i -particle index; k -discrete time index; v_i -velocity of i th particle; x_i -position of the i th particle p_i -best position found by the i th particle(personal best); G -best position by swarm (global best, best of personal bests) $g_{(1,2)}$ -random numbers on the interval $[0,1]$ applied to the i th particle.

The basic and necessary steps of customized PSO which are implemented in this study are as follows.

The initial population consists of solutions in the form of set of features, F , consisting of nine features X_i , $i = 1$ to 9.

$$F_i = \{X_1, X_2, X_3, \dots, X_9\} \quad (17)$$

The values of $X_i = 1$ if the feature is selected otherwise 0. F_i is one of the i th solution, out of total n solutions.

With each solution a numerical value of accuracy metric is associated, computed through machine learning classifier, which is to be maximized (that is objective function).

The fitness, in terms of accuracy parameter, is computed for both F_i and F_j , superior one will remain in race and weaker will be discarded.

The CPSO is repeated M number of times (termination condition) where

$$M = \text{multiply}(\text{number of images in the dataset, number of initial features in hand}) \quad (18)$$

Elitism is also applied in the CPSO, by retaining the best solution found so far. The best found retained solution, after the termination condition is satisfied, will be returned.

PSO is customized to implement self-crafted functions and variables (Refer Table 1) which are as follows. The values of the parameters used in the CPSO of features are revealed in Table 2.

Pseudo code of CPSO:

INPUT:: Two Feature Vectors consisting of 09 Best Scored Features which are returned by ‘univariate’ and ‘feature importance’ method respectively.

OUTPUT:: Optimized subset of features generating best Accuracy.

```

1. Begin
2. Step1: Produce the initial population using Initialize_Binary (num_population).
3. Step 2: Evaluate each individual (particle) in swarm.
4. Step 3: cal_pop_fitness (num_population, features, labels, train_data, test_data)
5.         Step 3.1 Store the binary value from particle into array.
6.         Step 3.2 Implement Feature Reducing Procedure//Line number 7 to 14
7.         for i in Length (num_population):
8.             Initialize rf as rf = []
9.             call reduce_features (i, features)
10.            Select all indices where value is equal to 1 in a particle
11.            Add into rf
12.            return rf
13.         Step 3.3: Evaluate each individual particle using different machine learning classifiers.
14.         return classifiers_accuracy
15. Step4: PSO operates on the population of particles to evolve the best solution
16.     while (not termination_condition) do
17.         begin
18.             Step 4.1:
19.                 Compare fitness value for each individual of swarm
20.                 Compute PBest(n) and GBest
21.                 Begin(Perform PSO operations)
22.                     Compute V(n + 1) from Equation (15)
23.                     Compute X(n + 1) from Equation (16)
24.                 End
25.                 Set n = n + 1
26.             Step 4.2 Repeat Step 3
27.         end
28.     end_while
29. Step 5: return the best subset of feature.
30. return (best_subset and best_accuracy)

```

TABLE 1 Description about self-crafted functions and variables used in CPSO

1. **cal_pop_fitness**: The fitness value for all the solutions is calculated by calling this fitness function.
2. **Input**: This function accepts the current population, the extracted features, the class labels, the train indices, and the test indices.
3. **Train indices**: Few samples are stored in training with their indices stored in train indices.
4. **Test indices**: Test samples indices are stored in test indices.
5. **cal_pop_funtion**: calculate the fitness value of all the solutions(particles).
6. **num_population**: it randomly creates an initial population.
7. **Reduce_features**: **Input**: num_population, features.
Output: return reduced feature vector.

TABLE 2 CPSO parameters

Parameter	Value
w	0.7
c1	1.5
c2	1.5
Maximum generation	(number of images in the data set * number of initial features)
Population size	50

The time needed to perform a particular operation is referred to as computational cost (or time complexity or computational time), and its measurement is similarly critical as other vital parameters. While computational cost may apply to a variety of computer resources (memory, power supply, etc.), it is most often used to refer to calculation time. The time complexity of CPSO's is primarily determined by two factors: (1) random initialization and (2) particle velocity and position updates. These two components can be written as $O(m \times N)$ in Big O notation, where m denotes the population and N denotes the problem dimensions is based solely on the time complexity of updating particle velocity and location. Although the CPSO has a lower computing cost than the fundamental PSO technique due to the absence of inertia weight update and confidence term calculation, the Big O notation may also be represented by $O(m \times N)$. According to the Big O notation, it retains an $O(m \times N)$ time complexity. In general, CPSO's complexity does not rise by orders of magnitude. The average convergence time for the CPSO-based KNN is ~ 398 s, which is much smaller than the time required for the other classifiers. Additionally, the CPSO-based KNN achieves a high level of classification accuracy.

4 | RESULTS AND EXPLORATION

As discussed, the proposed system (Figure 9) incorporates four models in itself. Thus, this section is bifurcated into four Subsections 4.2–4.5 and particularly each subsection is dedicated to depict the computed results from particular model. Six performance evaluation metrics have been preferred on the suggested prediction models whose brief description is given below. Two tables are presented depicting the features selected through univariate method and feature importance method.

4.1 | Performance measurement metrics

Using the Confusion Matrix, a machine-learning problem can be classified into two or more categories (Refer Table 3). Four distinct sets of predicted and actual values are represented in Table 3; 2 X 2 matrices are typical of confusion matrices. There are four possible outcomes in the uncertainty matrices: true positive (TP), false positive (FP), true negative (TN), and false negative (FN). It has been determined that a number of normative performance assessment metrics have been computed in this study. Four words, TP, FP, TN, and FN, form the basis of these. Patients with a disorder and a positive test result are said to have TP, whereas patients without a disorder and a positive test result are said to have FP. FN is a disease patient, but the test is negative for TN.

Figure 10 depicts a confusion matrix in which negative (0) represents normal and positive (1) represents glaucoma. Sensitivity (or recall) (a true positive rate, the number of accurate outcomes, the number of true positives separated by the sum of the true positives and false negatives) is one of the best measures of diagnostic efficiency. The evaluation shows us the frequency at which the classifier accurately predicts glaucoma. The specificity measure determines just how many of the negative cases were correctly predicted. Precision is how close to the target the sample comes. The word F1-score tests the accuracy of a source. The method incorporates the accuracy and recall of the model in order to determine a single value, the

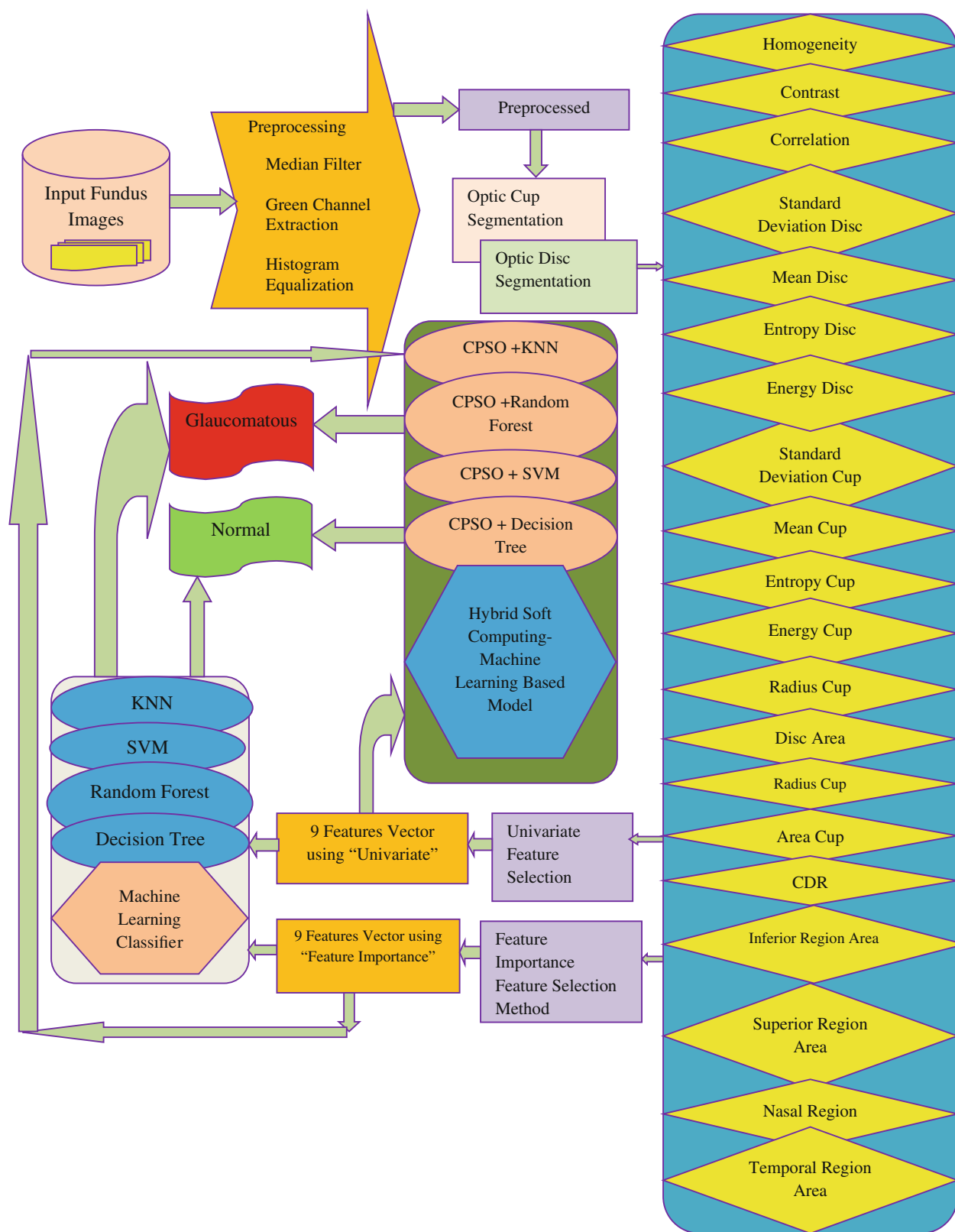
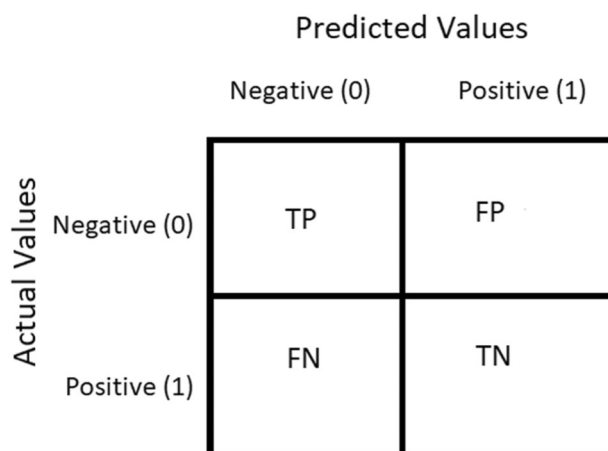


FIGURE 9 Complete layout of the proposed system for efficient glaucoma identification

harmonic mean. Precision is the amount of significance the measurement has on the outcome. Accuracy is the most important metric used to measure the efficiency of machine learning classifiers. A receiver operating characteristic curve (ROC curve) maps true and false positives against true positives at differing degrees of classification. The ROC study shows a reliable calculation of glaucoma classification. The region under the ROC curve indicates

TABLE 3 Confusion matrix

	Predicted cases	
	Glaucoma	Healthy (non-glaucoma)
Actual cases		
Positive	True positive (TP)	False negative (FN)
Negative	False positive (FP)	True negative (TN)

**FIGURE 10** The confusion matrix used in the proposed work is represented graphically

the sensitivity of the measurement. The field under the curve is a partially-independent vector which explores how the model is able to discriminate between regular and glaucoma. Raising the True Positive rate would boost both the False Positive rate and the True Positive rate. AUC measures the entire 2D area underneath the entire ROC curve, from (0,0) to (1,1). It provides an aggregate measure of performance across all possible classification thresholds. It is desirable for two reasons: scale invariant and classification threshold invariant.

$$\text{Sensitivity} = \frac{TP}{(TP + FN)} \quad (19)$$

$$\text{Specificity} = \frac{TN}{(TN + FP)} \quad (20)$$

Accuracy: As a result of mathematical computation, Equation (21) used to determine the accuracy of the classifier model.

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (21)$$

F1-Score: The Equation (22) used to compute the F1-Score.

$$F1score = 2 * \left(\frac{(\text{precision} * \text{recall})}{(\text{precision} + \text{recall})} \right) \quad (22)$$

Precision: When it comes to precision, it can be defined as the number of true positives divided by the total number of true positives plus the total number of false positives.

$$\text{Precision} = \frac{TP}{(TP + FP)} \quad (23)$$

Kappa: It refers to instruments that are used to assess the reliability of categorical items between and among raters. It assesses the degree to which a prediction corresponds to a true class. If there is no agreement among the raters other than what would be expected by chance, Kappa = 0, then the value of 1.0 represents complete agreement.

TABLE 4 09 features (along with their score values) returned by univariate feature selection

Features	Scores
CDR	213.656831
Entropy cup	134.128951
Mean cup	131.111267
Energy cup	129.21167
SD cup	112.561378
Area cup	107.135804
Energy disc	13.086123
Inferior region area	12.076122
Mean disc	11.085123

TABLE 5 09 features (along with their score values) returned by function importance feature selection

Features	Scores
Mean cup	0.202291
Area cup	0.171363
Entropy cup	0.170963
SD cup	0.079132
CDR	0.074013
Energy cup	0.059037
Entropy disc	0.030205
Inferior region area	0.030010
Mean disc	0.025218

$$p = \frac{(a+d)}{(a+b+c+d)} \quad (24)$$

$$ps = \left(\frac{(a+b)}{(a+b+c+d)} \right) * \left(\frac{(a+c)}{(a+b+c+d)} \right) \quad (25)$$

$$pn = \left(\frac{(c+d)}{(a+b+c+d)} \right) * \left(\frac{(b+d)}{(a+b+c+d)} \right) \quad (26)$$

$$pe = ps + pn \quad (27)$$

$$kappa = \frac{(p-pe)}{(1-pe)} \quad (28)$$

a = true positives, b = false positives, c = false negatives, d = true negative.

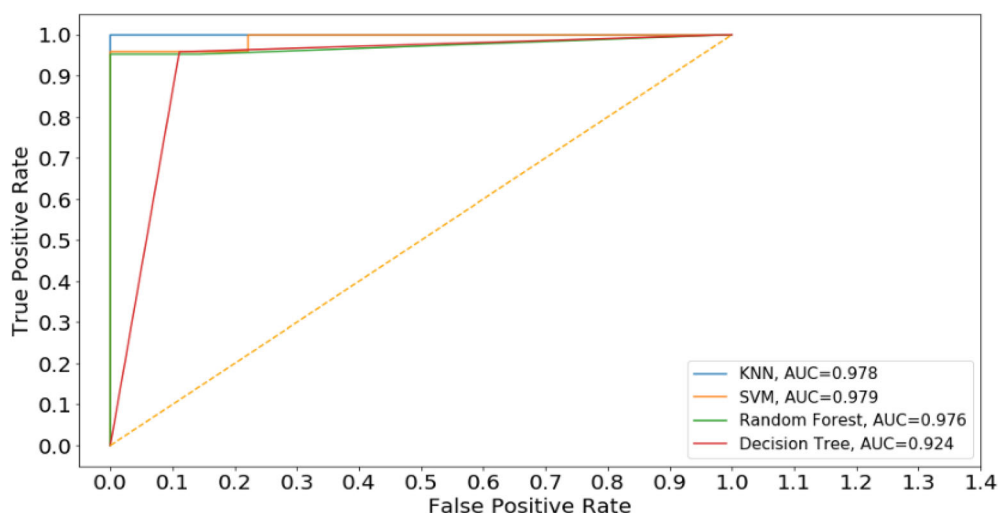
As discussed in the previous section, 20 vital features have been extracted from fundus photographs of public data set DRIONS-DB. Instead of working on all extracted features two separate functions have been applied to select the best 09 features (45% of the initial) from the initial set. The best features provided by these two methods along with their score values are presented (Tables 4 and 5).

4.2 | Performance evaluation with nine features selected through univariate method and forwarded to machine learning classifiers

In this case, the univariate selection approach returns nine attributes, which are then passed on to four shortlisted machine learning classifiers, and the resulting findings are seen in Table 6. Six indicators are used to compare the efficiency of these models. Accuracy and sensitivity is both equal for KNN, Random Tree, and SVM. When it comes to specificity and kappa, KNN comes out on top with 0.95 and 0.92 scores, respectively.

TABLE 6 Experimental results of different models using univariate

Models	Sensitivity	Specificity	Precision	F1-score	Kappa	Accuracy
KNN	0.95	0.95	0.97	0.97	0.92	98%
Random forest	0.95	0.93	0.92	0.92	0.89	93%
Decision tree	0.86	0.91	0.98	0.89	0.76	98%
SVM	0.95	0.90	0.90	0.92	0.85	98%

**FIGURE 11** ROC curve using univariate feature selection**TABLE 7** Experimental results of different models using CPSO with univariate features

Models	Sensitivity	Specificity	Precision	F1-score	Kappa	Accuracy	Running time
KNN	0.97	0.96	0.97	0.97	0.94	99%	398 (s)
Random forest	0.97	0.93	0.94	0.96	0.91	98%	410 (s)
Decision tree	0.89	0.92	1.0	0.93	0.82	99%	405 (s)
SVM	0.97	0.90	0.92	0.94	0.88	98%	401 (s)

In terms of precision, the decision tree performs well with a score of 0.98, led by KNN with a score of 0.97. The F1-value and Kappa score are both higher in the case of KNN and lower in the case of the decision tree. Figure 11 shows a ROC curve that has been drawn and displayed. It can be deduced that KNN is the highest performing variant in this situation.

4.3 | Performance evaluation with nine features selected through univariate method which are forwarded to CPSO-machine learning based sub-model

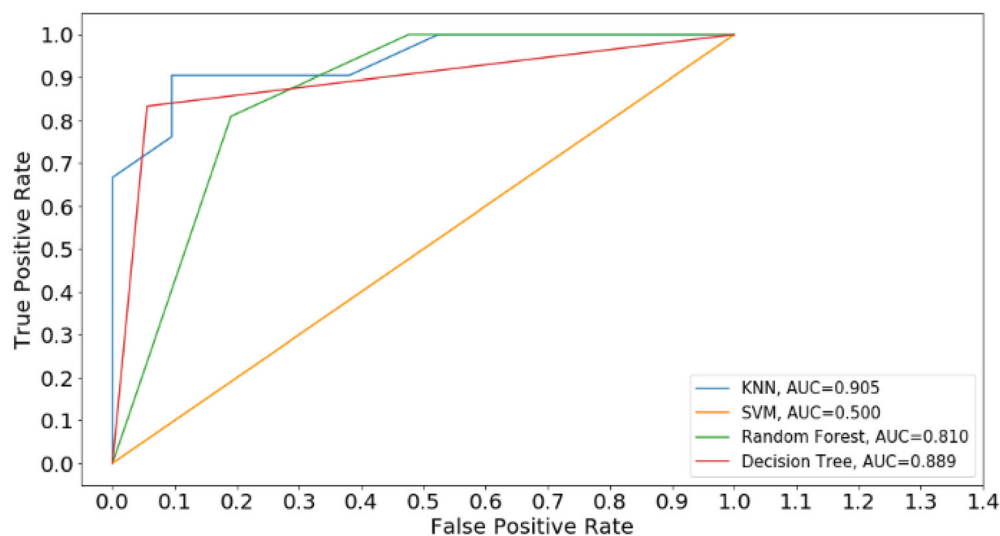
The 09 features returned by the univariate approach are fed into a hybrid model composed of CPSO and a machine learning classifier in the proposed framework's second model. The CPSO seeks an optimized subset of features, and machine learning-driven classifiers classify the images into two classes (glaucoma and healthy) based on the subset of features given by the CPSO. Due to the presence of four machine-learning classifiers, four options are generated: CPSO-KNN, CPSO-Random Forest, CPSO-Decision Tree, and CPSO-SVM. The four sets of results are provided by these four options and are shown in Table 7. Sensitivity scores for KNN, random forest, and SVM are all equal. KNN has the highest sensitivity, F1-Score, and kappa, ratings of 0.97, 0.97, and 0.94, respectively. With a score of 1.0, the decision tree produces the highest precision. Although the accuracy produced by the three models is approximately equal (99%), when all six performance assessment metrics are considered, it can be inferred that KNN outperforms all competitor models in the race. As part of the result review, the code is written in such a way that one can also figure out which features from CPSO are the most accurate. The number of features and their names are shown in Table 8.

TABLE 8 Table depicting optimal subset of features generated through CPSO

Models	Number of features	Features	Accuracy
KNN	3	CDR, SD cup, entropy cup	99%
SVM	3	CDR, area cup, energy cup	97%
Random forest	4	CDR, mean cup, energy cup, energy disc	98%
Decision tree	4	CDR, entropy cup, mean cup, and mean disc	99%

TABLE 9 Experimental results of different models using feature importance

Models	Sensitivity	Specificity	Precision	F1-score	Kappa	Accuracy
KNN	0.92	0.90	0.91	0.94	0.85	90%
Random forest	0.90	0.91	0.92	0.94	0.87	81%
Decision tree	0.68	0.90	0.97	0.90	0.65	88%
SVM	0.70	0.70	0.65	0.60	0.45	63%

**FIGURE 12** ROC curve using feature importance selection method

4.4 | Performance evaluations with nine features selected through feature importance method and forwarded to machine learning classifiers

In this third scenario, 20 features on hand have been fed to the feature importance method; this method returns the 09 most significant features (on the basis of scores), which are forwarded to 04 machine learning classifiers turn by turn, and the generated results are presented using Table 9. The best accuracy in this case is produced by the KNN classifier, which is 90%, while in the same scenario where 09 features were returned by the univariate method, an accuracy of 98% was obtained using KNN. Thus, it is concluded that on this public and well-accepted data set, the results generated by all seven metrics, via KNN, using the features returned by the univariate are better than the results generated by the features returned by the feature importance function. The same is also replicated in the case of random forest, decision tree, and SVM while considering almost all efficiency metrics. The ROC curve is also drawn and presented with the help of Figure 12.

4.5 | Performance evaluations with nine features selected through feature importance method which are forwarded to CPSO-machine learning based model

The nine features shortlisted by the feature importance function are redirected to the next stage in this final model. Following that, CPSO uses these features to find the optimized subset from this collection using four machine-learning classifiers. A mixture of CPSO and the four classifiers is used to produce the four choices. As a consequence, four sets of results are produced from these combinations, which are summarized in

TABLE 10 Table depicting optimal subset of features generated through CPSO

Models	Number of features	Features	Accuracy
KNN	3	Mean disk, entropy disc, and mean cup	95.4%
SVM	3	Area cup, energy cup, and entropy cup	97.8%
Random forest	3	Mean cup, entropy disc, and SD cup	97%
Decision tree	4	CDR, mean cup, entropy cup, and SD cup	99%

TABLE 11 Experimental results of different models using CPSO with feature importance method

Models	Sensitivity	Specificity	Precision	F1-score	Kappa	Accuracy	Running time
KNN	0.94	0.93	0.94	0.94	0.88	95.4%	430(s)
Random forest	0.97	0.93	0.94	0.96	0.91	97%	410(s)
Decision tree	0.85	0.92	1.0	0.93	0.84	99%	405(s)
SVM	0.97	0.90	0.92	0.94	0.88	97.8%	409(s)

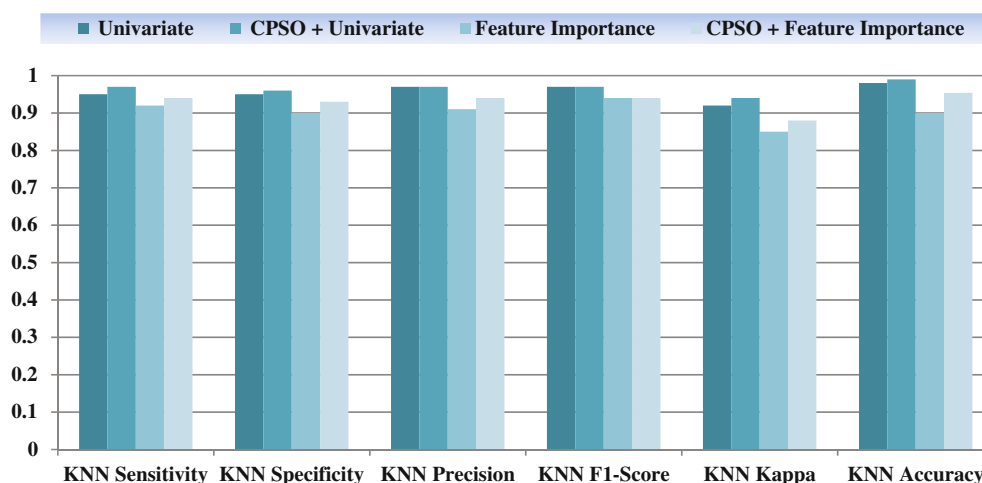
**FIGURE 13** Showcase of performance of KNN classifier on six metrics computed in four proposed model

Table 11. Random forest and SVM reach the greatest sensitivity. KNN and random forest produce the highest specificity. The decision tree has a maximum attainable precision of 1.00. In terms of sensitivity, random forest and SVM performed better. The random forest algorithm produces the best F1 and Kappa scores. In terms of accuracy, the decision tree and SVM are in first and second place, respectively. If one can examine this table holistically, with all six efficiency parameters in mind, Random Forest provides the highest performance but falls short of producing the best in terms of accuracy, which is the primary concern. The decision tree has the best accuracy but performs average in terms of sensitivity and Kappa metrics. While equating KNN and decision trees, the decision tree produces marginally higher accuracy and precision than KNN, but KNN is more promising in the other five metrics.

In addition, as seen in Table 10, decision trees need four features to generate results, while KNN needs only three features to generate results. To help with the preceding statement, code is written in Python in such a way that one can also decide which subset of features provides the best accuracy, as generated by CPSO; the number of features, along with their names, is displayed in Table 10. As discussed, one of the prime concerns is with the accuracy of the model followed by other efficiency metrics. The best accuracy generated through both of the models (univariate + CPSO + KNN and feature importance + CPSO + decision tree) is 99%. However, classifiers and feature selection methods are different. In the first case, it is KNN (Table 8), and in the latter case, it is decision tree (Table 10). Other than this, if one talks about remaining parameters and compares the results shown with the help of Tables 7 and 11, it can be concluded that on all these efficiency metrics, the simulated results produced by the four machine learning classifiers are superior in the case of the univariate method in comparison to the feature importance method. That is, in the case of SVM, the results of univariate+CPSO+SVM are more promising than feature importance+CPSO+SVM, and this is applicable to other classifiers also. Therefore, the authors will say that the CPSO-KNN model with the univariate feature selection method is the best model and that it can be used to find early signs of Glaucoma. The graphical representation of the performance of the machine learning classifiers in all four cases is shown in Figures 13 to 16.

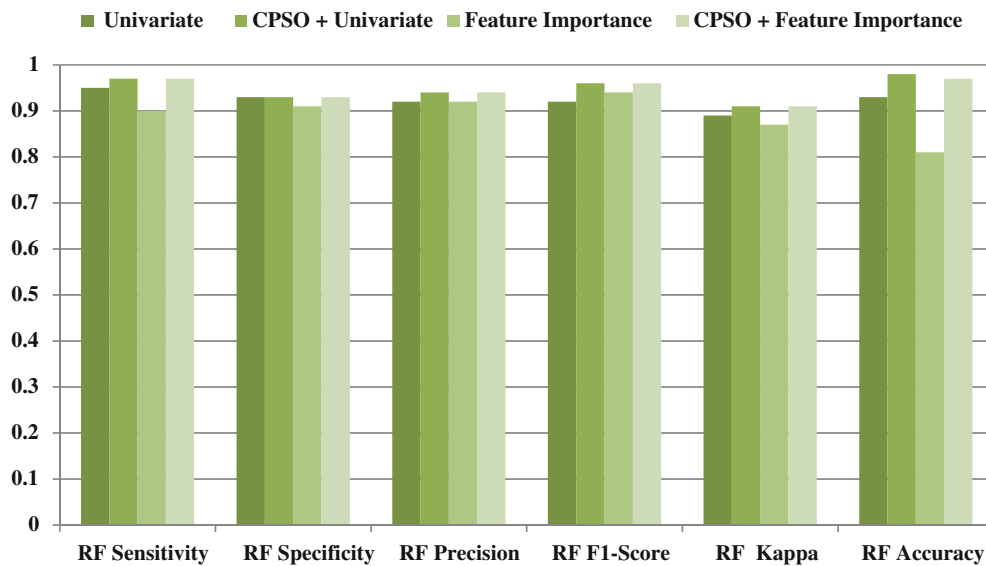


FIGURE 14 Showcase of performance of random forest classifier on six metrics computed in four proposed models

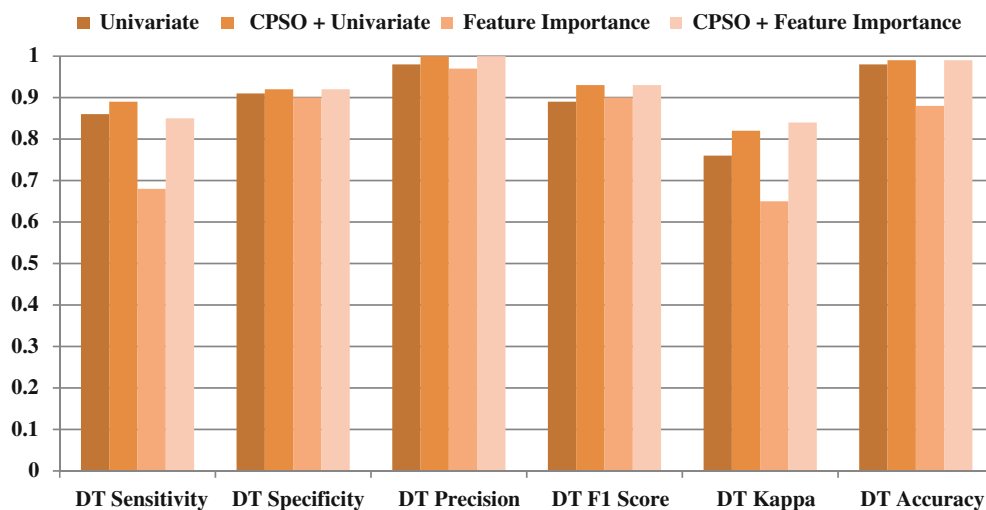


FIGURE 15 Showcase of performance of decision tree classifier on six metrics computed in four proposed models

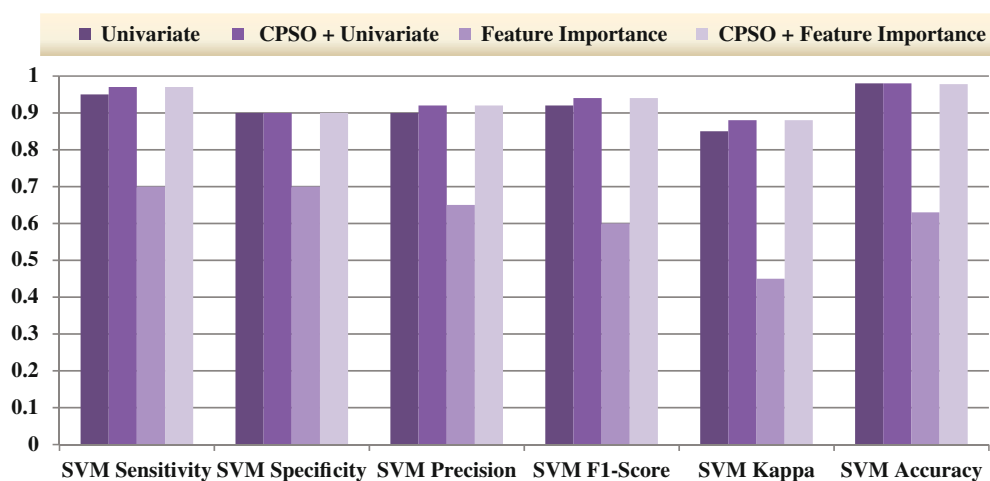


FIGURE 16 Showcase of performance of SVM classifier on six metrics computed in four proposed models

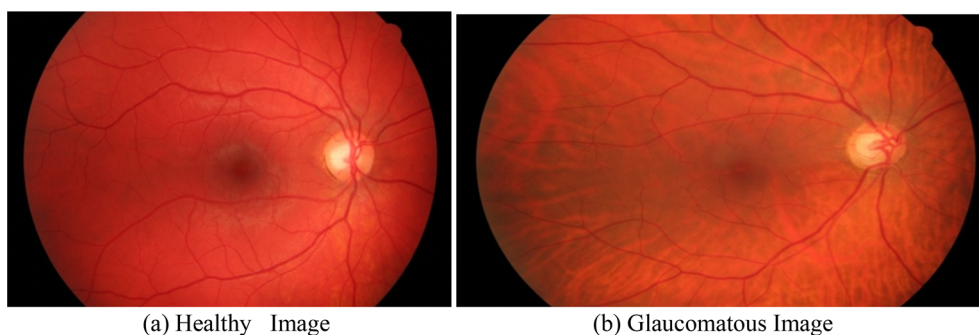


FIGURE 17 (a) Healthy image (b) glaucomatous image

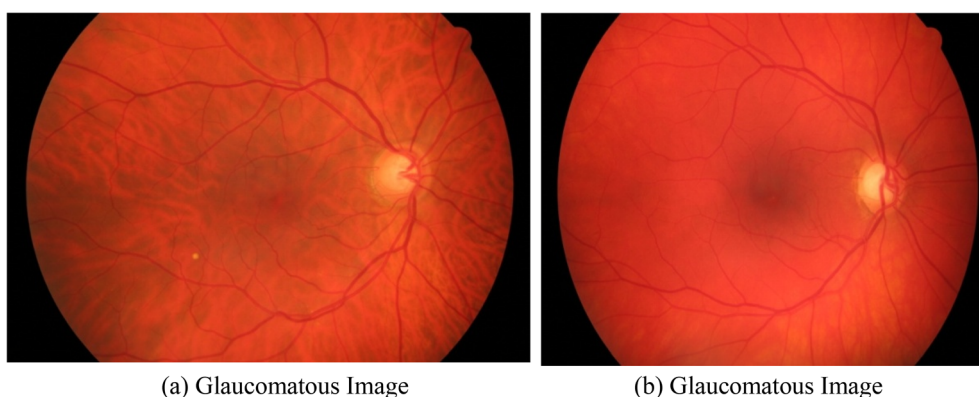


FIGURE 18 (a) Glaucomatous image (b) glaucomatous image

4.6 | Success and failure cases

In this sub-section, four scenarios are used to concentrate on the classification of the glaucoma images given. Two pictures from Healthy class are seen in Figure 17. Figure 17a reflects the situation. It is of a healthy (normal) subject since the picture is natural and usually expected by the proposed method. Figure 17b reflects the situation. Since the picture is glaucomatous and the recommended system predicts as usual (normal), it is a failure event. In the case of Figure 17b, it can be found that a greater part of the optic disc is filled by the optic cup, which is, of course, a characteristic of glaucomatous cases. With the aid of two photos of Glaucoma and the perpetrator class, Figure 18 illustrates two situations. Figure 18a presents a case of success, and Figure 18b presents a case of failure. If these statistics are compared, it can be found that the optic disc has a smaller optic cup in Figure 18b, which is a characteristic of a safe image in most cases.

4.7 | Comparison with related works

According to the data in Table 12, it is easy to conclude that the proposed work achieves the highest accuracy (prime concern). Maximum studies measured up to three precision parameters, while the proposed one shows the effects of six performance steps, suggesting that the proposed work was conducted after thorough research and review. Despite carefully reviewing the values of the parameters produced in the proposed scenario, it is possible to infer that the proposed approach has a fair balance between the metrics, meaning that both glaucoma and normal photos can be easily classified (refer Table 12). It should be noted that the data sets used in the studies listed in Tables 12 and 13 may differ.

The sensitivity level of Acharya et al. (2015) is lower than the suggested method. This shows the proportion of accurately classified instances of glaucoma and a missed case, which cost the patient, the chance of early treatment in real-life scenarios. Al-Bander et al. (2017) presented less impressive results on the evaluation criteria compared with the suggested method in their study. Acharya et al. (2011) provided a model for early and accurate diagnosis of glaucoma in the next study with 90% accuracy, which is somewhat less than in this empirical study. Gajbhiye and Kamthane (2015) used wavelet for the decomposition of photographs; these images were then used for the further extraction of characteristics. The experiments were simulated using wavelet and geometric moment characteristics on a large standard RIM-ONE r2 data set, and 90% accuracy was recorded in less computational time. The structure for the study of the texture of the optical disc image area was suggested by de Sousa

TABLE 12 Comparison table of selected prior techniques for automated glaucoma detection

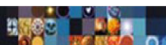
Work	Techniques	Sensitivity	Specificity	Precision	F1-score	Kappa	Accuracy
Acharya et al., 2015	Gabor features	89.75	96.2	–	–	–	93.1
Gajbhiye & Kamthane, 2015	Wavelet transform	–	–	–	–	–	86.57
de Sousa et al. (2017)	Geostatistical functions	95.0	88.2	–	–	–	91.2
Maheshwari et al., 2017	VMD, entropy	–	–	–	–	–	95.19
Al-Bander et al., 2017	CNN	85	90.8	–	–	–	88.20
Acharya et al., 2011	HOS	–	–	–	–	–	90.00
Khan et al., 2013	ISTN and CDR extraction	90.67	93.33	–	–	–	92.00
Das et al., 2016	Identifying CDR and NRR area in retina images	92.59	94.47	–	–	–	93.85
<i>Proposed</i>	Univariate + CPSO + KNN	97	96	97	97	94	99

TABLE 13 Comparison table of exclusive hybrid model based latest prior studies for automated glaucoma detection

Work	Techniques	Data set	Sensitivity	Specificity	Precision	F1-score	Kappa	Accuracy
Araújo et al. (2019)	Diversity Indexes	RIM-ONE	92.83	93.69	–	–	–	93.41
Krishnamoorthi & Chinnababu, 2019	SFFS and SVM, Naive Bayes and logistic regression	RIMONE	–	–	–	–	–	91.50
Jerith & Kumar, 2020	GW-NN	–	91.67	94.11	91.67	91.67	–	93.10
Perumal & Dhanasekaran, 2020	MSD and hybrid neural network	STARE and MESSIDOR	93.34	94.12	95.8	–	–	93.45
David & Jayachandran, 2020	HCSD with HRKSVM	Private Data Set	90	98.67	–	–	–	97.55
Pal et al., 2018	Customized deep learning multi-model	Combination of DRIONS-DB and three other data sets	–	–	–	–	–	AUC of 0.923
<i>Proposed</i>	Univariate + CPSO + KNN	DRIONS-DB	97	96	97	97	94	99

et al. (2017) for the identification of glaucoma, using the local binary pattern to represent the optical disc region and geostatistical functions to identify texture patterns. For classification purposes, the texture characteristics obtained were forwarded to SVM. Maheshwari et al. (2017) employed the Relief method for the selection of a private data collection of 488 images, which was selected for the experiment, and 95.19% accuracy was presented for the simulation.

Finally, an attempt has also been made to compare the proposed work with other recently published literature exclusively on 2-layered/3-layered implemented models for early glaucoma detection in reputed journals. A comparative table is presented at the end for readers (Table 13). The non-morphological features from the spatial domain are forwarded to the sequential floating forward (SFFS) Selection process, and the features returned from this selection method were forwarded to SVM, Naive Bayes, and Logistic Regression classifiers in the case of Krishnamoorthi and Chinnababu (2019). The proposed study is a move away from this work since CPSO has been used for feature optimization, which returns an improved subset of features. The obtained findings demonstrate the supremacy of the proposed work. The analytical study suggested by David and Jayachandran (2020) selects essential features using a hybrid colour and form descriptor (HCSD). Following that, SVM is used to build a hybrid model, and the best result is seen in the case of HCSD with Hybrid Radial Basis Kernel-centred Support Vector Machine (HRKSVM), which is 97.55%. Many variations may be identified as this work is related to the proposed one. The main problem is the data collection; the authors experimented with a private data set, while the suggested analysis uses a well-accepted mainstream public data set. In the implementation, the two processes, CPSO and major function collection, are absent. The utility is measured using just three metrics; the obtained precision is higher in the case of David and Jayachandran (2020), whereas the suggested method yields more positive results with all the remaining parameters. In the work proposed by Araújo et al. (2019), a feature selection protocol was introduced with the aid of GA, and then SVM was used to classify fundus eye images as glaucomatous or stable. The suggested approach achieves 93.41% accuracy, 92.83% sensitivity, and 93.69% specificity. When this method is opposed to the proposed work, there are several angles that show the supremacy of the latter. Initially, none of the approaches was used to shortlist the important features; nevertheless, two methods were used



in this situation. The findings are stronger and more promising as a consequence of these approaches than the previously described strategy (Araújo et al., 2019). The authors focused on a single machine learning-based classifier, but current work implements four. At the same time, six medical effectiveness measurement criteria have been computed, while the writers computed just three. Finally, the suggested solution produces more promising findings in all indicators.

In their early diagnosis experimental study, Perumal and Dhanasekaran (2020) suggested an image highlight portrayal technique called a multi-structure descriptor (MSD), which uses changes of pixels together to represent neighbourhood space structure details, including image extraction. Three other classifiers are used in addition to the hybrid neural network for classification. Both the important function selection approach and the soft-computing approach were not applied in comparison to the proposed work. If the collection of features had been streamlined, the produced results may have been improved. To end the comparison, the four performance parameters were calculated, as is common in other situations. The findings are stronger in the presented situation. The Grey Wolf Neural Network (GW-NN) was introduced in the final comparative hybrid model analysis by Jerith and Kumar (2020), whose findings were contrasted with NN, Nave Bayes, and SVM on various performance parameters. The Grey Wolf algorithm (similar to CPSO in the presented case) is used for optimization, and a neural network is used for classification. On the other hand, the effects are coherent and balanced, but they do not outperform the findings of the proposed solution. The writers, on the other hand, did not present the related details of the data set chosen for experiments; if the data set was public, private, or personalized, this information was not reported. Finally, the analysis (Jerith & Kumar, 2020) removes the important function selection process, which is present in the proposed one. When looking at Table 13, it is clear that the suggested work computes the greatest number of performance measurement metrics, however, the data set may be different. Furthermore, the developed findings are reliable, balanced, and superior to other competitive approaches; the proposed method supports its real-time applicability as a computer-aided glaucoma diagnostic method for the community, which aids in the effective identification of this disease.

5 | DISCUSSION

This section will describe in detail the real (conventional) challenges associated with glaucoma detection that inspired the development of a new computer-aided automated medical decision support system. Later in this part, the relevance of feature selection is discussed in detail, since it has been done twice in this study, initially using a statistical technique and then using nature-inspired computing approaches. Following that, study of the findings takes place and ties them to the structure of all algorithms. Hence, this part is devoted to in-depth examination from a variety of angles.

According to leading ophthalmologists, the best approach to curing glaucoma is to discover it early and treat it aggressively. Due to the absence of symptoms, some physicians have called this condition the 'snake thief of sight'. Glaucoma is a deadly condition caused by an imbalance in the intraocular pressure within the eye. It may result in the sufferer losing their sight for the rest of their lives. If diagnosed early enough, the disease's course may be delayed or even stopped. As a consequence, early detection is already the scientific community's primary objective. Glaucoma pre-screening via regular eye examinations becomes critical in this circumstance. Manual examinations of the eyes are prone to mistakes and may be fairly time-consuming since detecting glaucoma lesions early and accurately is a difficult job owing to their variable texture, shape, size, imperceptible border, complicated screening method, and high computational complexity. A glaucoma specialist will photograph the patient's retina (a 'coloured fundus picture') to determine the presence of glaucoma symptoms. Manual localization of glaucoma lesions requires skilled personnel to extract minute features from coloured fundus pictures and grade them according to a sophisticated grading system. Manual activities and inspection of coloured fundus pictures (or any other medical image) are time-consuming and subjective. All of this analysis involves human intervention, making it susceptible to human error and intra-observer variability. Ophthalmologists may also perform other glaucoma tests involving the amount of fluid in the eye, the angle of the drainage or outflow channel, and a visual examination of the retinal fundus image. Tonometry, pachymetry, and other intraocular pressure measurement techniques are often employed by ophthalmologists as diagnostic tools. Due to the subjective nature of these criteria and the possibility of human error (intra-observer variability), depending on the medical practitioner's expertise, there is an immense need for highly qualified doctors, who are in limited supply. Simultaneously, it is a lengthy operation that typically takes between 20 and 45 min to complete. Additionally, photographs may be evaluated by board-certified or specialist ophthalmologists to validate the accuracy of the diagnosis. There are various circumstances in which an ophthalmologist may wish to study a large number of pictures that do not present any problems. As a result, practitioners in the area of medical imaging are attempting to address these issues by analysing retinal images for the purpose of diagnosing glaucoma. As a consequence of this endeavour, professionals will be overworked, necessitating an automated solution based on a machine (such as a computer). Automatic scanning methods may overcome these obstacles by sorting through and recognizing glaucoma photographs and informing just those that are identified. Computing technology has advanced significantly in recent years, including more sophisticated components and sophisticated image processing algorithms. As a consequence, ophthalmologists need a computer-aided diagnosis (CAD) system to help them while working with medical pictures. This motivates us to propose a study on computer-assisted automated glaucoma diagnosis using retinal fundus photographs. To solve the aforementioned obstacles, machine-learning algorithms integrated with CAD automated systems for classifying retinal images as 'healthy' or 'infected' may be utilized.

To begin this genuine and innovative effort to locate glaucoma, the input images of a well-known, publicly accessible benchmark data set are pre-processed to remove any noise and objects present in them. Thus, on a publicly available standard database, the proposed system is qualified, examined, and evaluated. Twenty qualities (features) of these images (photos) are retrieved. These characteristics (features) are directed independently to two statistical procedures to form the vector that serves as input to the proposed structure in one way or another. The purpose of applying a statistical approach is to reduce the dimensions (feature selection) of the feature vector. The 'curse of dimension' is one of the most significant issues affecting the quality of optimization in the majority of data mining, pattern recognition, and machine learning jobs. Using high-dimensional data sets to train a classification model may impair the trained model's generalization ability. Additionally, the data set's high dimensionality leads to substantial computational and memory expenses. Feature selection (or reduction of features) is a critical data preparation technique in a wide variety of practical application fields relevant to expert and intelligent systems. The objective of feature selection (FS) is to extract a subset of interesting and relevant characteristics from a large feature collection. Thus, prior to the learning process, it is critical to treat the original data using a feature selection strategy to improve performance on the classification objective. FS methods are crucial in developing a system for predicting glaucoma infection. These algorithms are mainly focused on reducing redundant attributes from the original feature set in order to boost the prediction power of the classifier. The construction of such a system necessitates the use of a training set with relevant characteristics. As a result, prior to classifying, relevant characteristics with a high discriminative potential are chosen. The prediction abilities of classification techniques vary greatly depending on the feature selection. Optimization of features is a vital step in the development of data mining and machine learning systems. It improves the efficiency of classifiers and minimizes the amount of processing space required. Although selecting the most critical qualities from a large data set is a challenging task, FS approaches are used to discover the optimal set of features for the classifier. As mentioned before, 20 characteristics are taken from each subject's fundus picture, including the ISNT area (refer Figure 7). Rather than dealing with all the extracted features, two distinct methods have been used to choose the best nine characteristics from the original set. The univariate feature selection approach is used in Table 4 to identify the best nine characteristics. It operates by using a univariate statistical test to choose the best attributes. Each attribute is compared to the target variable to see whether or not they have a statistically significant relationship. Finally, all test scores were compared, and the traits with the highest scores were chosen. The chi-squared test is used as the statistical test parameter, with k equal to nine (the number of top features to pick). The chi-squared test is a technique for selecting univariate features. Table 4 summarizes the top-scoring nine attributes where CDR received the highest score of 213.656831. This demonstrates the importance of CDR traits in determining results.

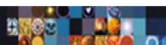
The feature importance selection approach is also used in parallel in Table 5 to identify the best 09 characteristics. The decrease in node impurity weighted by the probability of reaching that node is used to calculate feature importance. The node probability is calculated by dividing the number of samples that arrive at the node by the total number of samples. The greater the value, the more important the feature is. It computes node importance for each decision tree based on Gini importance. In the process of computing the feature importance, the following Equations (29), (30), and (31) are used. In these equations, k represents the significance of a node. The weighted number at node k is the impurity value. Left() is the left part of node k and Right() is the right part of node k . The value of each feature on a decision tree is then calculated as, where shows the importance of any node. The importance value is normalized between 0 and 1. The norm is the normalized value that is obtained by dividing the sum of all feature importance values.

$$pl_k = x_k C_k - x_{\text{left}(k)} C_{\text{left}(k)} - w_{\text{right}(k)} C_{\text{right}(k)} \quad (29)$$

$$fl_i = \frac{\sum_{k: \text{node } k \text{ splits on feature } i} pl_k}{\sum_{k \in \text{all nodes}} pl_k} \quad (30)$$

$$\text{norm} fl_i = \frac{fl_i}{\sum_{k \in \text{all features}} fl_k} \quad (31)$$

The kernel of this suggested system is based on an innovative mix of a modified variation of the well-known PSO (CPSO) algorithm with cutting-edge machine learning classifiers. The features chosen by the CPSO algorithm are utilized for the prediction of this infection using four classifiers. Due to the fact that the system operates in four distinct configurations, four distinct sets of derived results are compared using various efficiency metrics (like accuracy). There are four scenarios in all, and each scenario provides a unique set of outcomes. In the first scenario, the univariate selection strategy yields nine characteristics (Table 4), which are subsequently fed to four selected machine-learning classifiers. Six metrics are used to assess the efficacy of these models. For KNN, Random Tree, and SVM, both accuracy and sensitivity are equivalent. With scores of 0.95 and 0.92, respectively, KNN is the clear winner in terms of specificity and kappa. Precision-wise, Decision Tree outperforms KNN with a score of 0.98, while KNN outperforms Decision Tree with a score of 0.97. Both the F1-value and the Kappa score are greater in the case of KNN than in the case of the decision tree. As may be determined, KNN is the optimal version in this case. In the second scenario, the 09 characteristics given by the univariate technique are input into the proposed framework's second model, which is a combined model built of CPSO and a machine learning classifier. The CPSO determines an optimum subset of features, and machine learning-based classifiers divide the pictures into two



classes based on the CPSO's subset of features. Machine learning classifiers CPSO-KNN, CPSO-Random Forest, CPSO-Decision Tree, and CPSO-SVM are all examples of CPSO-KNN. This means that there are four possibilities because there are four classifiers. These four alternatives provide four distinct sets of results, which are summarized in Table 7. KNN, Random Forest, and SVM all have identical sensitivity ratings. The sensitivity, F1-Score, and kappa values for KNN are 0.97, 0.97, and 0.94, respectively. The decision tree achieves the maximum accuracy with a score of 1.0. While the accuracy of the three models is nearly identical (99%), it is clear that KNN outperforms all rival models in the race when all seven performance evaluation parameters are considered. As part of the result review, code is produced in such a manner that one can establish the most accurate subset of CPSO features; the number of features and their names are shown in Table 8. In the third case, 20 features are sent to the feature significance method; this technique returns the nine most important features (based on their scores), which are then transmitted to four machine learning classifiers in turn, as shown in Tables 5 and 9. The best accuracy is obtained by the KNN classifier in this situation, which is 90%, while in the identical scenario, when nine characteristics are provided by the univariate technique; the model acquires a 98% accuracy using KNN. Thus, it can be concluded that for this publicly available and well-acknowledged data set, the results obtained for all seven metrics using KNN using the univariate's features are much superior to the results created using the feature significance function's features. The same is true for random forest, decision tree, and SVM when practically all efficiency measures are included. For training and testing purposes, a fivefold cross-validation approach has been employed.

In the last scenario, the nine features selected by the feature value function are sent to the next step of the final model. Following that, CPSO uses these attributes to discover the optimal subset of this collection through four machine-learning classifiers. Four choices are generated using a combination of CPSO and four classifiers. As a result of these combinations, four sets of results are generated, which are reported in Table 11. Sensitivity is best for random forest and SVM. The most specific models are KNN and random forest. The decision tree may achieve a maximum accuracy of 1.00. In terms of sensitivity, random forest and SVM scored better. The random forest technique generates the optimal F1 and Kappa values. In terms of accuracy, decision trees and SVM rank first and second, respectively. When this table is evaluated holistically, taking into account all six efficiency factors, it is observed that Random Forest achieves the maximum performance but falls short of the best accuracy, which is a major concern. While the decision tree has the highest accuracy, it falls short in terms of sensitivity and Kappa measures. There are some differences between KNN and decision trees when they are compared. The decision tree achieves a little more accuracy and precision than KNN, but KNN is better on all the other metrics. Additionally, as seen in Table 10, decision trees need four attributes to provide results, while KNN requires just three. To help with this, Python code is written in such a way that one can also figure out which subset of features has the best accuracy, as shown by CPSO: the number of features, along with their names. As discussed previously, this study's primary concern is model accuracy, followed by other efficiency metrics. KNN and a decision tree are used to classify and choose features in different ways. In the first case, KNN is used (Table 7), while in the second, a decision tree is used (Table 11). The best accuracy for both models is 99%. Other than that, if one compares the simulated results from the four machine learning classifiers to the results shown in Tables 6 to 11, one can say that the univariate method is better for all of these efficiency metrics. For SVM, the results of univariate + CPSO + SVM are more promising than the results of feature importance + CPSO + SVM, and this holds true for other classifiers as well. Ultimately, the authors will say that the CPSO-KNN model with a single-feature selection method is the best model and that it can be used for early glaucoma detection. Figures 13 to 16 show how well the machine learning classifiers did in each of the four situations.

The KNN-CPSO hybrid model with univariate characteristics is shown to be the best model, demonstrating its superiority over other commonly used approaches, and thus detects glaucoma more precisely in the input images. Additionally, the suggested strategy has a greater sensitivity value than any other solution provided in recent years by academics. This suggests that the proposed technique is more effective at identifying potentially beneficial events, which is the ideal scenario for medical research. As a consequence of the presented discoveries, the basic outcome of glaucoma diagnosis will continue to be simplified, allowing for better daily clinical glaucoma management. It would allow all patients globally to protect themselves against further eyesight loss via prompt and precise medical interventions. Because the highest level of accuracy reached is in line with standards set by the American Academy of Ophthalmology (AAO), the work is likely to be good enough.

6 | REAL APPLICATION OF THE PRESENTED WORK

Fundus photography is currently one of the most often performed imaging methods; it is simpler to analyse the characteristics of the retina (the internal surface of the eye) in stereoscopic fundus photographs rather than checking one's own eye. It provides a bird's-eye view of the retina (the internal surface of the eye) and allows doctors to make the most accurate diagnosis possible. This is a type of progressive imaging that is commonly utilized by ophthalmologists. The retina is made up of 10 semi-transparent layers, each of which serves a specific purpose in the visual perception process. Fundus photography provides a bird's-eye view of the inner limiting membrane, the uppermost layer, and the other underlying layers. Because retinal disorders typically originate in one layer of the retina before spreading to the other layers (e.g., the production of cotton wool spots in the nerve fibre layer), it is crucial to have a sense of depth when evaluating a fundus to make an accurate diagnosis. These images provide detailed information about the layers of the retina. Traditionally, fundus cross-sectional photographs of retinal tissues were utilized to diagnose and treat eye problems. In important thickness measurement sectors for glaucoma diagnosis in the general population, fundus technologies have proven a comparable ability to differentiate between normal and glaucomatous eyes. The use of suggested three-layer approach on

fundus images based on statistics, nature-inspired computing, and machine learning (ML) has been shown to be efficient, accurate, and promising for glaucoma assessment. Together with previous study findings, our findings demonstrate the usefulness of a specific three-layered-based technique for identifying glaucoma in a real-world scenario. The results revealed that our technique is helpful in detecting this illness. The suggested three-layered phenomenon (Univariate + CPSO + KNN) on the subject fundus images has been shown to be efficient, accurate, and promising in assessing glaucoma. The suggested model outperforms current techniques in six efficiency assessment parameters (accuracy, sensitivity, precision, specificity, F1-score, and Kappa score), as shown in the preceding tables (accuracy, sensitivity, precision, specificity, F1-score, and Kappa score). To the best of our knowledge, no previous researchers have calculated all of these efficiency-related factors in a single study. The initial recovered attributes are likewise drastically reduced by more than 75%, to the lowest count. Our findings in most vital medical efficiency measuring criteria accuracy reaches upto greater than 99%, resulting in relevant results comparable to those of senior (or expert) ophthalmologists. The suggested techniques improve robustness by lowering dependency on picture quality and the impact of additional artefacts. By a wide margin, the suggested approach surpassed earlier state-of-the-art processes. As a result, we believe that our suggested approach will aid in the accurate prediction of glaucoma utilizing a fundus picture. The method described here is not limited to glaucoma; rather, it can be used to diagnose a wide range of ocular illnesses, particularly those involving the optic nerve head. This approach may be extended to a number of eye disorders by training on disease-specific data and making minor adjustments. The suggested model (Univariate + CPSO + KNN) has the potential to improve general eye health and speed the identification of blinding eye disorders in collaboration with human specialists. As previously established, the system's operating time (training or testing time) for delivering the final screening result for a single fundus picture is likewise rather excellent. Our proposed human-centred health care recognition system displays competency and accountability while also being socially acceptable. As a result, it is capable of operating autonomously in remote hospitals. The proposed technique could be used to detect glaucoma in remote areas where ophthalmologists are scarce, with only true cases referred to experts for further investigation. The proposed technology might be incorporated in mobile devices or linked to IoT-enabled devices, and it could be used to help people all over the world or in locations where there are not enough skilled eye doctors or detection equipment.

7 | CONCLUSION

Glaucoma is a catastrophic condition caused by an imbalance in the intraocular pressure within the eye. It may result in the sufferer losing their sight for the rest of their lives. Effective screening methods rely on specialists manually analysing the photos for illness recognition. However, due to the technical difficulties inherent in the screening approach and a scarcity of highly educated human resources (professional ophthalmologists), successful screening-oriented therapy is a costly endeavour. To address the limitations of a manual glaucoma diagnosis system, a robust automated medical decision support system embedding three distinct methods (statistical approach, nature-inspired approach, and machine learning techniques) has been established in the described work. Using a publicly available benchmark data set, DRION-DB, the suggested technique achieved up to 99% accuracy while reducing more than 75% of the originally extracted features. The given findings demonstrate that the presented technique is capable of properly determining the disease's adversity degree by displaying the corresponding historical examples for an appropriate prescription. While visual function is the primary concern for patients, progressive structural changes can occur prior to functional loss, and patients with more rapid fundus changes are at an increased risk of worsening visual loss, suggesting the possibility of escalating treatment earlier to preserve vision more effectively. In comparison to other recent glaucoma detection approaches, the proposed framework can effectively generate discriminative key points from low-intensity and noisy input pictures and categorizes them into their appropriate classes. As a result, this technique has the potential to significantly aid in the automated detection of glaucoma lesions and to support specialists in carrying out their duties more effectively. The suggested technique might be used to incorporate a computational tool for glaucoma detection in real-world circumstances. It may aid experts by providing auxiliary analysis during glaucoma identification, thus ensuring a higher quality of life and a more positive prognosis for glaucoma therapy. The ability to monitor glaucoma progression is likely to be enhanced further by novel strategies that incorporate information from the fundus and visual fields, thereby reducing the noise inherent in both tests, and such strategies are likely to be integrated into commercial devices in the coming years. Keeping an eye on how glaucoma might progress, using IoT devices, and connecting to the cloud are some of the things that will be important in the future.

Some shortcomings in this work have been recognized, which are as follows. Given the current focus on evidence-based medicine, it may not yet be appropriate to place complete reliance on machines. Including Macula scans and other publicly accessible features like intraocular pressure (IOP) and visual test readings would very certainly improve the method's accuracy further. This work is limited to a single standard data set. External validation of the suggested technique on bigger and more diverse data sets remains an option. Significant gains might be made by expanding the database's sample size. Additionally, the out-of-sample data set may be included to facilitate real-time implementation. For increased classification accuracy, advanced image processing and classification algorithms might be used. Additionally, other properties may be examined in addition to those stated in the suggested technique. Numerous recent soft-computing-based techniques are suggested, as well as their hybrid variants. Certain algorithms may be more efficient in terms of accuracy, feature reduction, or both. We have made the assumption that the patient has just one eye infection, if any. The presented efforts are restricted to categorization, regardless of whether the person has glaucoma or not. This work makes no distinction between the stages (mild, moderate, or severe) of glaucoma infection.

CONFLICT OF INTEREST

On behalf of all authors, the corresponding author states that there is no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no data sets were generated or analysed during the current study.

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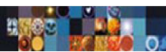
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